



Imperial Aeronautics

Hybrid Electric Transport

Venture

HYBRID ELECTRIC TRANSPORT

Individual Report

FATT 05: FLIGHT DYNAMICS AND CONTROL

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Abstract

The goal of the Hybrid Electric Transport project is to design an aircraft that is able to achieve fuel savings given its hybridisation. Through this project, not only were the feasibility, key technological requirements and potential cost savings explored but also the conceptual design of the aircraft. The conceptual design was distributed amongst sub-groups; this report's focus is on the methodology adopted to ascertain the stability of the aircraft as well as to determine if the aircraft was statically stable and able to fulfil CS-25 regulations. It was found that the aircraft is statically stable with the results being substantiated by the dynamic analysis, tail and control surface sizing.

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Nomenclature

The following list prescribed the symbols and conventions used in the report below. Standard flight dynamics conventions have also been adopted in the abbreviation in of the stability derivatives. The coordinate system assumed in this report is that of the stability axis, follows a right hand rule. Note that the longitudinal distances (along the fuselage) are in general with respect to the nose of the aircraft.

Acronyms

SHSS Steady heading sideslip

CG Centre of Gravity

Greek Symbols

α Angle of attack

α_p Angle of attack experienced by the propeller

α_{0_p} Thrust setting angle of the aircraft

β Side slip angle

β_p Sideslip angle experienced by the propeller

Δ Change in

δ Control surface deflection angle

δ_a Aileron deflection angle

δ_r Rudder deflection angle

δ_t Trim tab deflection angle

$\delta_{a,sat}$ Saturation aileron deflection angle

$\delta_{r,sat}$ Saturation rudder deflection angle

ϵ Wing twist at root

$\frac{\partial \epsilon}{\partial \alpha}$ Downwash gradient

γ Proportion of thrust produced by both turbo-props

ω Angular velocity

ϕ Bank angle

σ Ratio of propeller blade area t disc area

τ Control surface effectiveness parameter

ϵ Downwash

ϵ_s Sidewash

ϵ_{d0_p} Propeller downwash angle with fuselage reference line at zero angle of attack

$\epsilon_{s\beta}$ Sidewash gradient

Roman Symbols

$\mathcal{R}$ Aspect ratio of wing

$\overline{C_l}$ Local lift coefficient of propeller

$\bar{c}$ Mean aerodynamic chord of wing

$\overline{V_v}$ Vertical tail volume coefficient

A Area swept by the propeller blades

b Span

C_w Coefficient of weight

d_p Propeller diameter

G Gearing ratio

i_h Tail setting angle

J Propeller Advance Ratio

K_f Empirical constant which quantifies the effect of wing position on the fuselage

l Rolling moment

l_f Fuselage length

l_h Horizontal distance between the aerodynamic centre of the horizontal tail and the CG

l_h Horizontal distance between the aerodynamic centre of the vertical tail and the CG

M Mach number

n Yawing moment

N_p Normal force generated by turboprop

p Roll rate

r Yaw rate

S Area

SM Static margin

T Thrust

V Velocity

W/S Wing loading

w_f Fuselage maximum width

Y Side force

Z_t Vertical displacement of thrust line

Z_v Vertical distance between the aerodynamic centre of the vertical tail and the CG

Superscripts

' Stick free

Subscripts

0 At zero angle of attack

β Side slip derivative

a	Aileron	p	Roll rate derivative
ac	Aerodynamic Centre	r	Rudder
B	Turboprop	r	Yaw rate derivative
$c/4$	Quarter chord	S	Electric motor
d	Drag	s	Control surface
e	Elevator	T	Thrust
f	Fuselage	t	Trim tab
H	Hinge moment	TO	Take off
h	Horizontal tail	v	Vertical tail
L	Lift	w	Wing
mc	Minimum control	Y	Side force derivative
np	Neutral point	z	Normal force
p	Propeller		

1 Introduction

The nature of flight dynamics and control of the conceptual design of an aircraft is highly dependent on designs and decisions made by the propulsion, aerodynamics, structures and baseline configuration teams. Initial configuration warrants general design recommendations to respective teams to ensure stability of the aircraft. The only design constraint placed upon other teams is span restriction, ensuring ailerons experience clear flow. The rationale behind this decision will be further elaborated in Section 6.

2 Stability Methodology

To simplify the analysis, the main assumptions in the static stability analysis carried out are: the aircraft is rigid and experiences small perturbation; and the earth is flat. Removal of aforementioned assumptions will significantly increase complexity and provide a level of fidelity that is not required at the conceptual design stage.

In order to ascertain differences between the static and dynamic stability of a hybrid-electric aircraft compared to a conventional aircraft, contributions to and assumptions made for obtaining stability derivatives were examined. Findings were that the analysis would be identical to that of a turboprop aircraft with multiple engines, whereby the effect of a propeller on the pitch and yaw stability derivatives need to be accounted for. Methodology for this correction will be elaborated on in Section 3. To ensure the aircraft is stable, the methodology adopted is iterative, illustrated as a flow chart in Figure 1.

2.1 Tornado and Stability Derivatives

Since derivatives are based on the response in forces and moments due to a given perturbation [2], this will be highly dependent on lift forces and geometry of the aircraft. Therefore, Tornado 3D-vortex lattice Method (VLM) open source software was chosen to estimate derivatives. This software was chosen as validity of derivatives obtained have been assessed: Reference [3] which compares Tornado stability derivatives to flight test data of a Boeing 747; Reference [4] compares control derivatives obtained through DATCOM procedure and Tornado and Reference [5] compares stability derivatives of a Cessna 172 from flight test data to those obtained through Tornado. All references suggest that Tornado is able to provide a reasonable estimate of the stability derivatives. However, there are limitations due to the following assumptions. Firstly, small angles of attacks are assumed. Secondly, fuselage effects and friction drag are not considered. Lastly, effects of thickness of lifting surfaces and compressibility are neglected [6]. The first assumption will not have a significant impact as it is consistent with the small perturbation assumption made for this analysis.

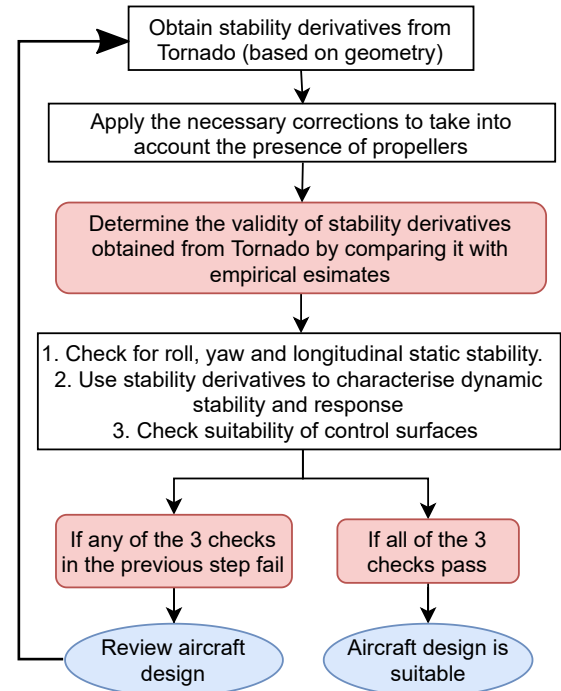


Figure 1: Flow chart demonstrating the iterative process of the methodology which was adopted to ensure stability

Remaining assumptions will result in certain derivatives not being able to fully describe the response of the aircraft. For example, fuselage has a substantial influence on the side force due to sideslip ($C_{Y,\beta}$) [5]. Consequently, empirical methods such as those detailed in ESDU and DATCOM were used to estimate stability derivatives that are vital to both dynamic and static stability of the aircraft.

The Tornado derivatives in Table 1 were obtained by Reference [1] in conjunction with the aerodynamics team. Figures A.1, A.2, A.3 and A.4 illustrate the model used in Tornado. Empirical estimates were calculated together with fellow members of the flight dynamics and control team. Both Tornado and empirical estimates in Table 1 have the propeller correction applied. Due to the tedious nature of obtaining empirical estimates and time constraint, stability derivatives were only found for cruise conditions, with particular focus on the lateral derivatives.

2.1.1 Roll Rate Derivatives

The p (roll rate) derivatives obtained by Tornado are in theory more accurate than calculated estimates - roll rate is mainly dependent on the lift generated by the wings. Features, such as fuselage, have not been modelled due to little contribution. Thus, these values can be trusted and have been used in static and dynamic analysis. Empirical estimates were calculated using

$$C_{Y_p} \approx (C_{Y_p})_v = \frac{2(Z_v \cos \alpha - l_h \sin \alpha)}{b_w} C_{Y_{\beta_v}}$$

Table 1: Comparison of stability derivatives (at cruise conditions) obtained using Tornado with empirical estimates along with corresponding deviations.

	Tornado	Estimates	Deviation
C_{Y_β}	-0.6003	-0.9871	64.4%
C_{Y_p}	0.0327	0.1034	216%
C_{Y_r}	0.5314	0.7920	49.0%
C_{l_β}	0.0835	-0.1302	55.9%
C_{l_p}	-0.6465	-0.7636	18.1%
C_{l_r}	0.1167	0.1143	2.06%
C_{n_β}	-0.2427	0.1726	28.9%
C_{n_p}	0.0323	0.0795	146%
C_{n_r}	-0.2173	-0.3180	46.3%
$C_{Y_{\delta_a}}$	0.0083	0	100%
$C_{Y_{\delta_r}}$	-0.2836	0.1875	38.8%
$C_{l_{\delta_a}}$	0.1812	0.1991	9.88%
$C_{l_{\delta_r}}$	0.0370	0.1678	354%
$C_{n_{\delta_a}}$	1.11E-3	5.27E-4	52.5%
$C_{n_{\delta_r}}$	-0.1220	-0.2112	73.1%

$$C_{n_p} = -\frac{C_L}{8}$$

$$C_{l_p} = \frac{4C_{L_{\alpha w}}}{S_w b_w^2} \int_0^{b_w/2} cy^2 dy$$

as detailed in Reference [7]. It was observed from Table 1 that these estimates are larger than those obtained through Tornado. This can be attributed to the lack of knock down factors such as taking into account the presence of 3D effects.

2.1.2 Sideslip Derivatives

β (sideslip) derivatives are significantly influenced by the presence of the fuselage. Thus, derivatives obtained through Tornado will not be as accurate as those obtained through empirical estimates. The Tornado value for C_{l_β} suggests that the aircraft is statically roll unstable. This is unlikely as design decisions made in the initial sizing were made to ensure this, further illustrating Tornado's inability to produce accurate results for derivatives that are heavily influenced by the presence of fuselage.

C_{Y_β} includes contributions from the wing, vertical tail and fuselage. Various components were estimated using graphs in Reference [8]

$$C_{Y_{\beta v}} = -\frac{\eta_v S_v C_{L_{\alpha v}}}{S_w} \varepsilon_{s_\beta}$$

where ε_{s_β} is the sidewash gradient. In general, C_{l_β} is affected by the dihedral, wing sweep, fuselage, tail, propellers and flaps [9]. Since the aircraft has no sweep

and dihedral, these effects are ignored. A high mounted wing results in a change in $C_{l_\beta} = -0.006$ 1/deg [10]. The tail contribution is given by

$$C_{l_{\beta v}} = -\eta_v \frac{S_v Z_v}{S_w b_w} C_{L_{\alpha v}} (1 - \varepsilon_{s_\beta})$$

$$C_{l_{\beta h}} = 0.08 \eta_h \frac{S_v S_h}{S_w b_w} C_{L_{\alpha h}} (1 - \varepsilon_{s_\beta})$$

Propellers have a destabilising effect: a positive sideslip which results in positive rolling moment. This effect is exacerbated when flaps are deployed. However, these effects are neglected as they are small [9]. The fuselage, wing and propellers contribute to C_{n_β} . Contribution from the wing is primarily due to the presence of sweep [11]. This has been ignored due to the aircraft wing being unswept. The effect of the propeller and fuselage have been quantified using

$$C_{n_{\beta p}} = 4 \left\{ \frac{d_{pB}^2 x_1 C_{N_{pB,\alpha}}}{S_w b_w J_B^2} (1 - \varepsilon_{s,\beta})_p + \frac{d_{ps}^2 C_{N_{pS,\alpha}}}{S_w b_w J_s^2} (x_2 + x_3) (1 - \varepsilon_{s,\beta})_p \right\}$$

$$C_{n_{\beta f}} = C_{m,\alpha_f} \frac{\bar{c}_w}{b_w}$$

where propeller effects are elaborated on in Section 3. A full derivation is found in Appendix C.2.

2.1.3 Yaw Rate Derivatives

Based on empirical estimates for r (yaw rate) derivatives from Reference [7],

$$C_{n_\beta} = -2C_{Y_{\beta v}} \frac{l_v}{b_w}$$

$$C_{n_r} = -2C_{L_{\alpha v}} \eta_v \bar{V}_v \frac{l_v}{b_w}$$

$$C_{l_r} = \frac{C_L}{4} - 2 \frac{l_v Z_v}{b_w^2} C_{Y_{\beta tail}}$$

Since derivatives are mainly dependent on the contributions from the vertical tail and fuselage, Tornado results and empirical estimates should largely be in agreement with each other. However, Table 1 shows that this is not the case with the exception of C_{l_r} . This could potentially be due to the way in which the vertical tail has been modelled as in the case of Reference [5], which compares results for a Cessna 172. However, due to time constraints on the project, this aircraft was not explored. In the same fashion, values used were those of the empirical estimates - it was not possible to determine validity of results from Tornado.

2.1.4 Control Derivatives

$C_{l_{\delta_a}}$ and $C_{Y_{\delta_a}}$ were estimated using Reference [7] which suggests that $C_{Y_{\delta_a}} = 0$ and

$$C_{l_{\delta_a}} = \frac{2C_{L_{\alpha w}} \tau_a}{S_w b_w} \int_{x_1}^{x_2} cy dy \quad (1)$$

where x_1 is the inboard aileron location and x_2 is the outboard aileron location. The Tornado result for $C_{l_{\delta_a}}$ is largely in agreement with the estimates. This is mainly due to lift effects. Equation 1 is an approximation that uses the concept of strip theory to determine $C_{l_{\delta_a}}$. τ_a , taking into account how lift distribution changes with a given deflection angle. Tornado will be able to provide a more accurate result as it is not based solely on historical data but rather actual geometry and forces acting on the model. Thus, the value obtained from Tornado was used. $C_{n_{\delta_a}}$ was estimated by taking the derivative of Equation 5 with respect to δ_a at the cruise angle of attack. The theory and assumptions will be elaborated on in Section 2.2.

Reference [8] estimates $C_{n_{\delta_r}}$ and $C_{Y_{\delta_r}}$ using

$$C_{n_{\delta_r}} = -\eta_v \tau_r \bar{V}_v C_{L_{\alpha_v}}$$

$$C_{Y_{\delta_r}} = \eta_v \tau_r \frac{b_r}{b_v} \frac{S_v}{S_w} C_{L_{\alpha_v}}$$

As with yaw rate derivatives, Tornado and empirical results should be in agreement with each other as contributions come mainly from the tail (which was modelled). However, Table 1 shows they are not. $C_{l_{\delta_r}}$ was estimated by taking the derivative of Equation 4 with respect to δ_r at the cruise angle of attack. The theory and assumptions will be elaborated on in Section 2.2.

2.2 Isolated Control Surfaces

Main assumptions made in this analysis is that control surfaces are flat-plate surfaces and are isolated from any bodies i.e. behaviour is not affected by the presence of fuselage or other lifting surfaces. However, these assumptions implies that results will not be accurate in all flight phases [12]. As such, it was chosen due to simplicity and availability of the method and is intended to be used as a sanity check, a means to gain an understanding of deflections required for control surfaces, and to obtain particular control derivatives that were challenging to acquire by other means.

The effect of an increment in elevator deflection on the normal force acting on the control surface at a given angle of attack and the effect of an increment in rudder deflection on the side force at a given angle of attack can be related as follows [12],

$$\frac{\Delta C_z}{2} = \frac{S_s}{S_w} \left[\text{sgn}(\alpha + \delta) \sin^2(\alpha + \delta) \cos \delta - \text{sgn} \alpha \sin^2 \alpha \right] \quad (2)$$

$$\frac{\Delta C_Y}{2} = \frac{S_s}{S_w} \frac{\text{sgn} \delta \tan^2 \delta \cos^2 \alpha \cos \delta}{1 + \tan^2 \delta \cos \alpha} \quad (3)$$

For ailerons, roll and yaw are coupled resulting in the

following equations:

$$\frac{\Delta C_l}{2} = \frac{S_s}{S_w} \cos \delta [\text{sgn}(\alpha + \delta) \sin^2(\alpha + \delta) - \text{sgn}(\alpha - \delta) \sin^2(\alpha - \delta)] \quad (4)$$

$$\frac{\Delta C_n}{2} = \frac{S_s}{S_w} \sin \delta [\text{sgn}(\alpha + \delta) \sin^2(\alpha + \delta) + \text{sgn}(\alpha - \delta) \sin^2(\alpha - \delta)] \quad (5)$$

Figure 2 and 3 are plots of Equation 4. Plots of Equations 2, 3 and 5 have been included in Appendix B.

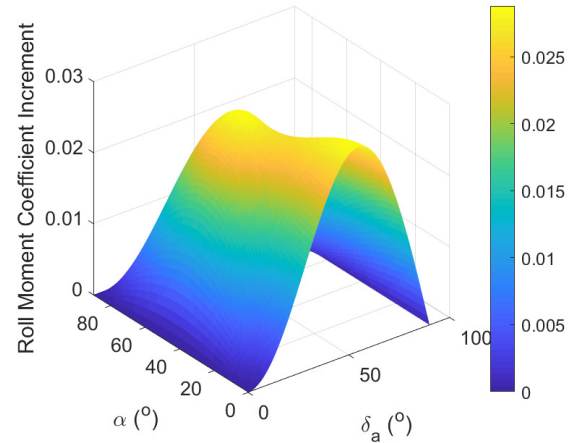


Figure 2: 3D plot of roll moment increment due to an aileron deflection and change in angle of attack

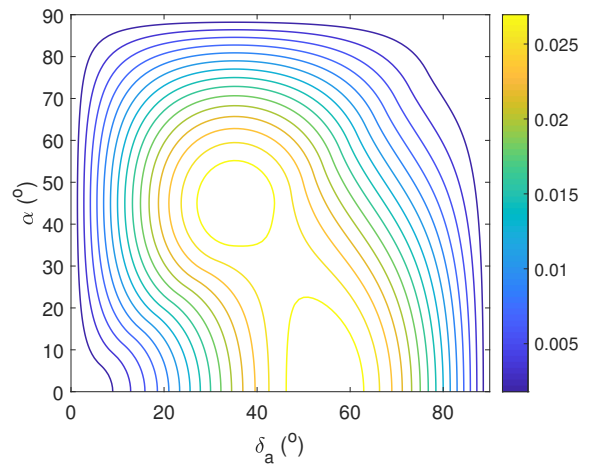


Figure 3: Contour plot of roll moment increment due to an aileron deflection and change in angle of attack

This theory was used to approximate $C_{l_{\delta_r}}$ and $C_{n_{\delta_a}}$ by taking the derivatives of Equations 4 and 5, using a central difference scheme in MATLAB at the trim angle of attack in cruise with respect to δ_r and δ_a respectively, to obtain control derivative estimates. Since these are secondary effects due to a given control surface deflection, the response of the aircraft will be dependent not only on the lift and the control surface but also the geometry and aircraft configuration (e.g. fuselage will produce a stabilising yawing moment for $C_{n_{\delta_a}}$ [5]). Since these effects are not modelled by either methods, none will provide accurate estimates. Aforementioned

assumptions made in this theory implies a relatively crude approximation that has been mainly attributed to control surfaces being modelled as flat plates instead of an extension of the aerofoil. As such, Tornado values for these derivatives were used in static and dynamic analysis.

3 Propeller Correction

From Reference [13], normal force generated by the propeller at an angle of attack, α , is given by

$$N_p = \frac{\sigma q A}{2} \left\{ \overline{C}_l + \frac{aJ}{2\pi} \ln \left[1 + \left(\frac{\pi}{J} \right)^2 \right] + \frac{\pi}{J} C_d \right\} \alpha \quad (6)$$

and the yawing moment is given by

$$n_p = -\frac{\sigma q A d_p}{2} \left\{ \frac{2\pi}{3J} \overline{C}_l + \frac{a}{2} \left(1 - \left(\frac{J}{\pi} \right)^2 \ln \left[1 + \left(\frac{\pi}{J} \right)^2 \right] \right) - \frac{J}{\pi} C_d \right\} \alpha \quad (7)$$

where

$$\overline{C}_l = \frac{3J}{2\pi} \left[\frac{2J}{\sigma q A \pi} T + C_d \right] = \frac{3J}{2\pi} \left[\frac{2S_w J}{\sigma A \pi} C_T + C_d \right] \quad (8)$$

By considering steady level flight, $\overline{C}_l$ can be found as thrust is equal to drag. Since the aircraft has more than one type of propeller, turboprop and electric motors, $\overline{C}_l$ will vary based on the proportion of drag it overcomes. Thus, by assuming the proportion of thrust produced by each propeller is used to overcome the same proportion of drag and defining γ as the proportion of total thrust produced by both turboprops with values obtained from [14]. The thrust can be related to the drag by

$$C_{TB} = \frac{\gamma C_d}{2} \quad C_{TS} = \frac{(1-\gamma)C_d}{4} \quad (9)$$

Substituting the relations in Equations 9 into Equation 8 results in

$$\overline{C}_{lB} = \frac{3J_B}{2\pi} \left[\frac{S_w J_B \gamma}{\sigma A \pi} + 1 \right] C_d \quad (10)$$

$$\overline{C}_{lS} = \frac{3J_S}{4\pi} \left[\frac{S_w J_S (1-\gamma)}{\sigma A \pi} + 1 \right] C_d \quad (11)$$

By non-dimensionalising Equation 6 and 7 and taking the partial derivatives with respect to the angle of attack,

$$C_{N_p, \alpha} = \frac{\sigma}{2} \left\{ \overline{C}_l + \frac{aJ}{2\pi} \ln \left[1 + \left(\frac{\pi}{J} \right)^2 \right] + \frac{\pi}{J} C_d \right\} \quad (12)$$

$$C_{n_p, \alpha} = -\frac{\sigma}{2} \left\{ \frac{2\pi}{3J} \overline{C}_l + \frac{a}{2} \left(1 - \left(\frac{J}{\pi} \right)^2 \ln \left[1 + \left(\frac{\pi}{J} \right)^2 \right] \right) - \frac{J}{\pi} C_d \right\} \quad (13)$$

From Reference [13], the yawing moment coefficient is linearly proportional to the propeller angle of attack,

α_p , such that

$$C_{n_p} = \frac{n_p}{\rho(\omega/2\pi)^2 d_p^5} = C_{n_p, \alpha} \alpha_p \quad (14)$$

It also gives that the side force coefficient is linearly proportional to the propeller sideslip angle, β_p , such that

$$C_{Y_p} = \frac{Y_p}{\rho(\omega/2\pi)^2 d_p^4} = C_{Y_p, \alpha} \beta_p \quad (15)$$

In the longitudinal case, the change in pitching moment about the centre of gravity is given by,

$$(\Delta C_m)_p = (\Delta C_{m0})_p + (\Delta C_{m, \alpha})_p \alpha \quad (16)$$

where

$$\begin{aligned} (\Delta C_{m0})_p &= -\frac{(1-\gamma)(z_2 + z_3)}{4c_w} C_d - \frac{z_1 \gamma}{2c_w} C_d + \\ &\quad \frac{2d_{pS}^2 C_{N_{S, \alpha}}}{S_w c_w J_S^2} (x_2 + x_3) (\alpha_{0p} - \varepsilon_{d0p}) + \\ &\quad \frac{2x_1 d_{pB}^2 C_{N_{B, \alpha}}}{S_w c_w J_B^2} (\alpha_{0p} - \varepsilon_{d0p}) \\ (\Delta C_{m, \alpha})_p &= \frac{2d_{pS}^2 C_{N_{S, \alpha}}}{S_w c_w J_S^2} (x_2 + x_3) \left(1 - \left(\frac{\partial \varepsilon}{\partial \alpha} \right)_p \right) + \\ &\quad \frac{2x_1 d_{pB} C_{N_{B, \alpha}}}{S_w c_w J_B^2} \left(\frac{\partial \varepsilon}{\partial \alpha} \right)_p \end{aligned}$$

where x_1 , x_2 and x_3 are defined in Figure C.1. The free body diagram and assumption made for this aircraft in the derivation can be found in Appendix C.1.

In the lateral case, the change in yawing moment about the centre of gravity is given by,

$$(\Delta C_n)_p = (\Delta C_{n0})_p + (\Delta C_{n, \alpha})_p \alpha + (\Delta C_{n, \beta})_p \beta \quad (17)$$

where

$$\begin{aligned} (\Delta C_{n0})_p &= \frac{4d_{pB}^3 C_{n_{pB, \alpha}}}{S_w b_w J_B^2} (\alpha_{0p} - \varepsilon_{d0p}) \\ (\Delta C_{n, \alpha})_p &= \frac{4d_{pB}^3 C_{n_{pB, \alpha}}}{S_w b_w J_B^2} \left(1 - \left(\frac{\partial \varepsilon}{\partial \alpha} \right)_p \right) \\ (\Delta C_{n, \beta})_p &= 4 \left\{ \frac{d_{pB}^2 x_1 C_{N_{pB, \alpha}}}{S_w b_w J_B^2} (1 - \varepsilon_{s, \beta})_p + \right. \\ &\quad \left. \frac{d_{pS}^2 C_{N_{pS, \alpha}}}{S_w b_w J_S^2} (x_2 + x_3) (1 - \varepsilon_{s, \beta})_p \right\} \end{aligned}$$

The free body diagram and assumption made for this aircraft in the derivation can be found in Appendix C.2.

Table 2: Change in stability derivatives due to the presence of propellers.

	Cruise	Take off	Landing
$(\Delta C_{n_\alpha})_p$	-0.00441	-0.00367	-0.00351
$(\Delta C_{n_\beta})_p$	0.00418	0.00381	0.00390
$(\Delta C_{m_\alpha})_p$	0.0121	0.0110	0.0113

Table 2 shows the change in stability derivatives for this aircraft due to the presence of propellers at a given flight condition. The values indicate that the propellers cause a stabilising effect in yaw but a destabilising effect in pitch which agree with Reference [15].

3.1 Criticism of Propeller Correction

The method simplifies the problem by disregarding the effect of propeller-induced velocity and assumes a propeller blade of constant chord. This implies that the change in direction of sectional lift force has been neglected [16]. Also, the normal force acting on the propeller, and thus the propeller's impact on the stability of the aircraft, is an underestimate.

The impact of the helical path of the induced velocities in the slipstream on the rest of the aircraft were also not taken into account. Considering the large number of propellers on this aircraft, it is likely that this will have a substantial impact on the wing and vertical tail of the aircraft. In future iterations, this effect should be quantified and integrated into not only the stability analysis but also the empennage design.

Depending on the geometry of the propellers and its relative position to the CG, at low speeds, the destabilising yawing moment contribution is relatively invariant with power but may have a considerable effect that increases with velocity. The change in rolling moment experiences noteworthy fluctuations with power, especially with flaps deployed. This could be attributed to the propeller effect which would increase the lift generated by the wing. These could potentially have an effect on the spiral and dutch roll modes of the aircraft [17]. Therefore, the impact of power of the propellers at the aircraft's range of velocities and the interaction with flaps is another key area to be looked into in future iterations.

A more accurate method which is able to predict the aerodynamic forces and moments (that is in agreement with experimental data) is to integrate Goldstein's vortex theory, blade element theory and Prandtl-Betz tip-loss approximation to predict induced velocity [16]. However, due to the level of complexity of this method and, although, propellers have a significant effect on the stability of the aircraft; at this stage, it will provide a level of fidelity that is not necessary. In addition, the availability of simpler methods such as those described in References [18], [19], [15] and [20] as well as the method mentioned in this section, meant that it was not necessary to explore this area in such depth. In addition, these methods and method used are accepted as valid methods [21]. Nonetheless, improving the approximation of these effects in conjunction with the propulsion team will be key in future iterations pertaining to the design of the aircraft.

4 Longitudinal Static Stability

General pitch stability criterion for any aircraft [9] is given by,

$$\frac{\partial C_m}{\partial \alpha} \equiv C_{m,\alpha} < 0 \quad (18)$$

this must be ensured for the aircraft which must be able to achieve trim in accordance to requirements from CS-25.

4.1 Methodology

Initially, it was found that the aircraft was longitudinally unstable whilst working in partnership with the baseline configuration team for the weight and balance of the aircraft [22]. The following approach was taken to stabilise the plane: firstly, fixing of location and initial sizing of tail, hence reducing the complexity by artificially reducing the number of free variables. Secondly, wing position was adjusted based on the relationship between horizontal tail area and wing position for the aftmost CG location in landing at a given static margin, as seen in Figure 5. The range of positions of the aerodynamic centre of the wing were constrained by landing gear positions to be between 11.5 and 12.9m [23] from the nose of the aircraft. The aerodynamics team also requested to ensure that the CG is between the wing and tail, resulting in both lifting surfaces producing positive lift, improving the aerodynamic efficiency of the aircraft. Moving the wing significantly changes the CG location, allowing longitudinal static stability to be achieved, based on the range CG locations on the ground and in cruise, pitch stability criterion, static margin and ability to trim were checked at cruise, take off and landing. If the aircraft is able to trim, pitch stability criterion was fulfilled and static margins are within limits suggested by literature (4 – 7% and not exceeding 20% [24]), the aircraft was deemed stable. Else, wing position is moved through the full range of possible locations. If not, the tail would be resized.

4.2 Stick Fixed Analysis

Stick fixed analysis was used to determine trimmability of the aircraft, fulfilment of pitch stability criterion and if the static margin is within limits suggested by literature. It was found that the aircraft was longitudinally stable for a wing position of 12 m from the nose without changes to the tail sizing. The CG locations are as shown in Table 3.

Table 3: CG location along length of fuselage with a wing position of 12 m, in cruise and on the ground.

	Foremost	Average	Aftmost
Cruise	12.73 m	12.87 m	13.00 m
Ground	12.82 m	12.95 m	13.08 m

Pitch stability was checked by considering rate of change of pitching moment with respect to a change in angle of attack when taking moments about the CG. It was found that

$$C_{m,\alpha} = -C_{L_{\alpha w}} \frac{x_{acw} - x_{cg}}{\bar{c}_w} + C_{M_{\alpha f}} + (\Delta C_{m,\alpha})_p - \eta_h C_{L_{\alpha h}} \frac{S_h}{S_w} \frac{x_{ac_h} - x_{cg}}{\bar{c}_w} \left(1 - \frac{\partial \varepsilon}{\partial \alpha}\right) \quad (19)$$

where the downwash angle with wing angle of attack is empirically approximated [21] by

$$\frac{\partial \varepsilon}{\partial \alpha} = 4.44(K_A K_\lambda K_h \sqrt{\cos \Lambda_{c/4}})^{1.19} \frac{C_{L_{\alpha w}}|_M}{C_{L_{\alpha w}}|_{M=0}} \quad (20)$$

where K_A , K_λ and K_h are empirical constants, and the coefficient of moment due to fuselage is empirically approximated [24] by

$$C_{m_{\alpha f}} = -K_f \frac{l_f w_f^2}{\bar{c}_w S_w} \quad (21)$$

Components of $C_{m,\alpha}$ that are contributed by the wing and tail corroborated with results from Tornado, thus validating the method and results obtained. Results of Equation 19 have been tabulated in Table 4, showing a fulfilment of the general pitch stability criterion.

Table 4: $C_{m,\alpha}$ at cruise, take off and landing conditions

	Cruise	Take off	Landing
$C_{m,\alpha}$	-0.33314	-0.31865	-0.24669

The static margin and trimmability of the aircraft was determined by considering pitching moments generated about the CG of the aircraft, given by,

$$C_m = -C_{L_w} \frac{x_{acw} - x_{cg}}{\bar{c}_w} + C_{M_{0w}} + C_{m_{\alpha f}} \alpha - \eta_h C_{L_h} \frac{S_h}{S_w} \frac{x_{ac_h} - x_{cg}}{\bar{c}_w} + \frac{Z_t T}{q S_w \bar{c}_w} \quad (22)$$

$$C_L = C_{L_w} + \eta_h C_{L_h} \frac{S_h}{S_w} \frac{x_{ac_h} - x_{cg}}{\bar{c}_w} \quad (23)$$

where [24]

$$C_{M_{0w}} = \left[C_{m_{0airfoil}} \right]_{M=0} \left(\frac{R \cos^2 \Lambda_{c/4}}{R + 2 \cos \Lambda_{c/4}} \right) - 0.01 \epsilon \left[\frac{C_{L_{\alpha w}}|_M}{C_{L_{\alpha w}}|_{M=0}} \right]$$

$$C_{L_w} = C_{L_{\alpha w}} (\alpha + i_w - \alpha_{0w})$$

$$C_{L_h} = C_{L_{\alpha h}} \left[(\alpha + i_w - \alpha_{0w}) \left(1 - \frac{\partial \varepsilon}{\partial \alpha}\right) + (i_h - i_w) - (\alpha_{0h} - \alpha_{0w}) \right]$$

The static margin is defined as [24],

$$SM = \frac{x_{np} - x_{cg}}{\bar{c}_w} \quad (24)$$

where x_{np} is given by Equation 45. The full derivation can be found in Appendix D.

To ascertain the relationship between static margin, location of the aerodynamic centre of the wing and the horizontal tail, the location of the wing and horizontal tail area was varied in Equation 45, resulting in a change in values obtained in Equation 24. This was then plotted in Figure 4. To obtain the relationship between horizontal tail area and wing location, an intersecting plane of a static margin of 7% was plotted in Figure 4 aftmost CG landing plot. The intersection line was then plotted in Figure 5, enabling one to go from one variable to another. Figure 5 suggests that a larger tail area is required for more forward wing location, agreeing with the logic that change in wing location results in a larger moment arm for lift generated by the wing. As such, the horizontal tail will be required to produce a larger lift force to balance this.

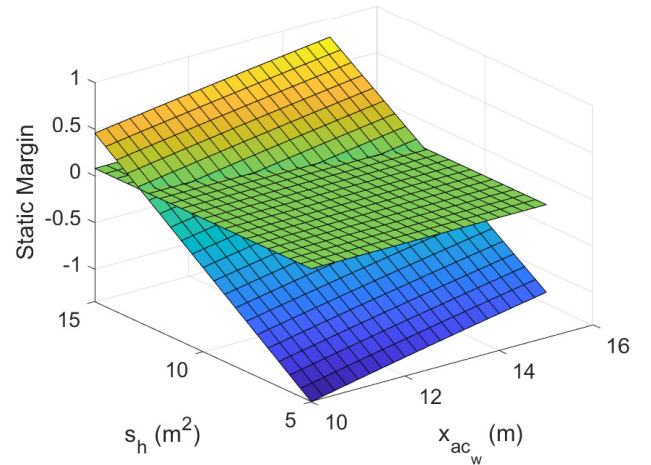


Figure 4: Relationship between static margin, horizontal tail area and wing position for the aftmost CG location during landing.

Plotting graphs for aforementioned CG locations and flight conditions established that the most constraining condition was at landing with aftmost CG location. This corroborates with results in Table 5, showing lowest static margin is at landing with aftmost CG location. Consequently, the plot shown in Figures 4 and 5 are for this condition.

Table 5 also suggests that the aircraft is sufficiently stable - the most stable configuration being below 20% [24] and cruise and take off configuration at the aftmost CG location being approximately within the recommended 4 – 7% [24]. The most constraining condition is not within the recommended limits.

The aircraft configuration can be further optimised in

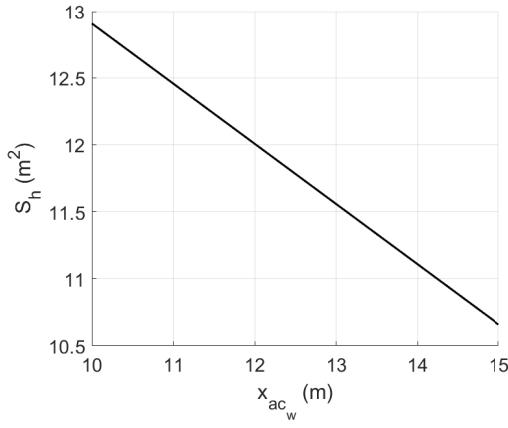


Figure 5: Relationship between horizontal tail area and wing position for the aftmost CG location during landing at a static margin of 7%.

Table 5: Static margin in critical flight conditions for the foremost, aftmost and average CG position

	Foremost	Average	Aftmost
Cruise	15.8%	10.8%	5.64%
Take off	16.8%	10.4%	3.96%
Landing	15.3%	8.86%	2.44%

future iterations by fully integrating tail sizing and location into this process along with merging the code with that of the weight and balance team. Figure 5 suggests that for the chosen wing position, required tail area is less than the current aircraft's tail area. Consequently, a smaller horizontal tail could potentially have been used instead, leading to weight savings due to reduced structures required, cost and improvement in the aerodynamic efficiency of the aircraft. However, reduced horizontal tail area might not be as great as expected, as Figure 5 shows the relationship without considering convergence with pitch stability criterion, weight and balance, and trimmability. Lack of convergence is associated with the discrepancy between static margin calculated and desired static margin for the most constraining condition.

To determine trimmability of the aircraft at foremost and aftmost CG location, trim plots were plotted for the average CG location. Lines for foremost and aftmost CG locations seen in Figures 6, 7 and E.1 represent where pitching moment coefficients are equal to zero at the given CG location, as the pitching moment at the plotted CG location is related to the coefficient of lift by,

$$C_m = C_L \overline{\Delta_{cg}} \quad (25)$$

where $\overline{\Delta_{cg}} = \frac{x_{cg2} - x_{cg1}}{C_w}$, x_{cg2} is the CG location of interest and x_{cg1} is the CG location for plotted data [25].

Figures 6, 7 and E.1 show that the aircraft is trimmable

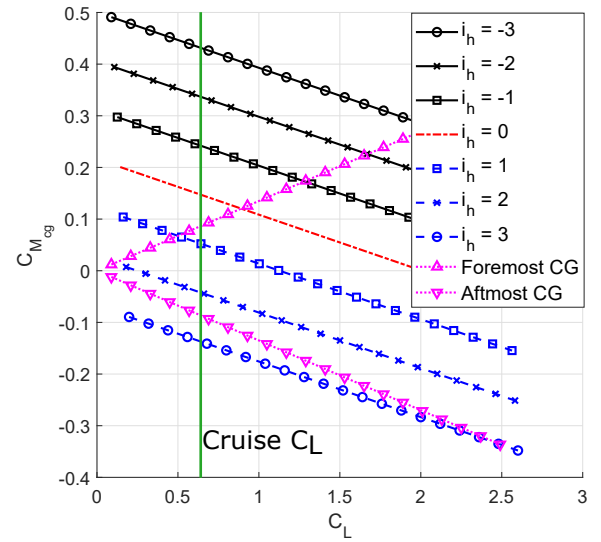


Figure 6: Trim plot for cruise at average CG location

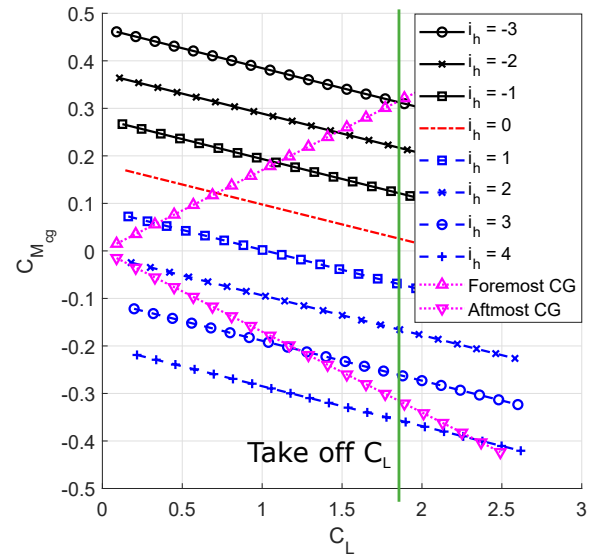


Figure 7: Trim plot for take off at average CG location

at all three configurations. However, the range of required tail setting angle is relatively large. Tail sizing could be a probable issue, though the cause has yet to be ascertained. This is an area that should be looked into in future iterations.

4.3 Stick Free Analysis

The governing equations for steady level flight are given by,

$$0 = C_{m0} + C_{m_\alpha} \alpha + C_{m_{\delta_e}} \delta_e \quad (26)$$

$$0 = b_1 \alpha_h + b_2 \delta_e + b_3 \delta_t \quad (27)$$

where $b_1 = \frac{\partial C_H}{\partial \alpha}$, $b_2 = \frac{\partial C_H}{\partial \delta_e}$, $b_3 = \frac{\partial C_H}{\partial \delta_t}$ and $\alpha_h = \alpha + i_h - \varepsilon = (1 - \frac{\partial \varepsilon}{\partial \alpha}) \alpha + i_h$. Since an adjustable tail was chosen, there are no contributions from δ_t term which is to take into account the effect of the trim tab.

In stick-free, elevator deflection is given by

$$\begin{aligned}\delta_e &= \delta_{e, float} = -\frac{b_1}{b_2}\alpha_h \\ &= -\frac{b_1}{b_2}\left(1 - \frac{\partial \varepsilon}{\partial \alpha}\right)\alpha + i_h\end{aligned}\quad (28)$$

Substituting Equation 28 into Equation 26, the required tail setting angle to achieve trim for a given angle of attack is given by,

$$i_h = \frac{b_2}{b_1 C_{L_{\delta_e}}}\left\{\left[C_{L_\alpha} + \frac{b_1}{b_2}C_{L_{\delta_e}}\left(\frac{\partial \varepsilon}{\partial \alpha} - 1\right)\right]\alpha - \frac{1}{q}\frac{W}{S}\right\}\quad (29)$$

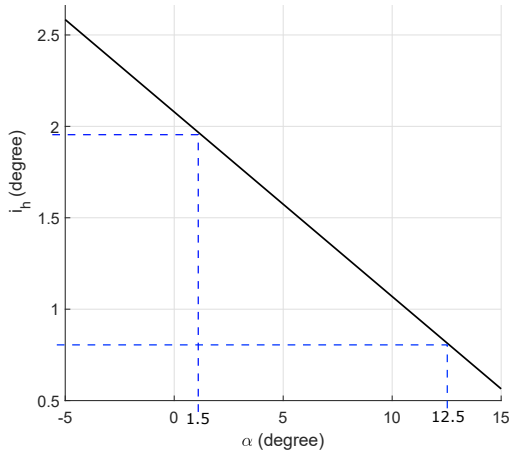


Figure 8: Stick free trim angle against angle of attack in cruise conditions at average cruise CG location.

This was to ensure that the aircraft is able to achieve trim within the range of tail setting angles alongside a level of agreement between stick fixed and stick free analysis. With reference to Figure 8, the former has been demonstrated to be true, showing that required tail setting angle ranges between 0.5° and 2.5° . The latter is also illustrated by Figure 8. At cruise angle of attack of 1.5° [26], stick free trim tail setting angle is approximately 2° , while stick fixed trim tail setting angle is between 1° to 2° (with reference to Figure 6). At $C_{L_{max}}$ which corresponds to an angle of attack of 12.5° [26], in cruise, stick free trim tail setting angle is approximately 0.75° , stick fixed trim tail setting angle is between 0° to 1° . Since the stick fixed and stick free neutral point are relatively close to each other (with reference to Table 6 and Figure 9), it is reasonable to presume that the required tail setting angles are at a high level of agreement with each other.

The stick free neutral point can be found by considering how a free floating elevator will cause a change in the lift and pitching moments of the aircraft. This has been quantified using

$$x'_{np} = x_{cg} + \frac{C'_{m_\alpha}}{C'_{L_\alpha}}\quad (30)$$

where $C'_{m_\alpha} = C_{m_\alpha} - \frac{b_1}{b_2}C_{m_{\delta_e}}(1 - \frac{\partial \varepsilon}{\partial \alpha})$ and $C'_{L_\alpha} = C_{L_\alpha} -$

$$\frac{b_1}{b_2}C_{L_{\delta_e}}(1 - \frac{\partial \varepsilon}{\partial \alpha})\quad [9].$$

Table 6: Distance between the nose and the stick free neutral point in cruise, take off and landing configuration.

	Cruise	Take off	Landing
Stick free neutral point (m)	13.10	13.01	13.00

4.4 Stick Force

Since reversible mechanical controls were used, it was important to determine stick force required to enact a change in the controls and ensure that it meets CS-25 requirements. It was assumed that the aircraft experiences a constant acceleration. As such, stick force was estimated using,

$$F = GS_e \bar{c}_e \frac{W}{S} \frac{C'_{L_\alpha} b_2 x_{cg} - x'_{np}}{\Delta} \left(1 - \frac{V}{V_{trim}^2}\right)\quad (31)$$

where gearing ratio (G) used being consistent with those used by Reference [27], $\Delta = -C_{L_\alpha} C_{m_{\delta_e}} + C_{m_\alpha} C_{L_{\delta_e}}$ and V_{trim} depends on which configuration one is trying to achieve trim in. Stick force gradients were found by derivating Equation 31 with respect to velocity,

$$\left.\frac{dF}{dV}\right|_{V_{trim}} = -GS_e \bar{c}_e \frac{W}{S} \frac{2}{V_{trim}} \frac{C'_{L_\alpha} b_2 x_{cg} - x'_{np}}{\Delta} \quad (32)$$

From Equation 31, it can be observed that as the velocity of the aircraft reaches trim velocity, force required decreases. In addition, when the aircraft reaches trim velocity, stick force required is zero. Furthermore, with reference to Equation 32, stick force gradient is largest when the difference between the CG and stick free neutral point is the greatest. Thus, values in Table 7 were obtained at the foremost CG position. Results obtained for cruise, take off and landing configurations are shown in Table 7. As cruise has the steepest gradient, this agrees with the rationale that since cruise is the highest velocity the aircraft is travelling at, control surfaces experience the largest force due to dynamic pressure. It also shows that stick force gradient does not exceed 1.3 Ns/m, fulfilling the requirements in CS-25.173 (c).

Table 7: Stick force gradient at cruise, take off and landing with the foremost CG position.

	Cruise	Take off	Landing
Stick force gradient (Ns/m)	1.23	0.561	0.532

4.5 CG Limits

For an aircraft with reversible controls, taking into account handling qualities of the aircraft, forward and aft limit of the CG are limited by the following, for aftmost CG location,

1. $x'_{np} < x_{np}$ [28]
2. The magnitude of control gradients are not exceedingly small [28]
3. For the aircraft to be stable with fixed controls, $x_{cg} < x_{np}$ and $x_{cg} < x'_{np}$ with free controls [21]

For the foremost CG location.

1. The stick force gradient must not exceed 4 N per 3.06 m/s [29] in trim conditions
2. Control force required to land must not exceed 356N at speeds from trim to approach speed [29]

Figure 9, alongside preceding analysis, shows that most of the above conditions have been fulfilled. Since the aircraft meets all requirements for longitudinal static stability, this was validated against the longitudinal dynamic stability [1] and was found to be in agreement.

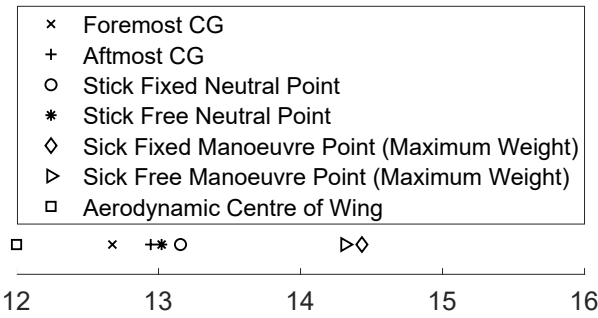


Figure 9: Longitudinal location of various points, in cruise, with respect to the nose.

In addition to preceding requirements, another key area that should be further looked into are stick force gradient per g, location of stick fixed and stick free manoeuvre points [21]. These would determine how sensitive the aircraft is to a control input that could potentially result in structural failure of the airframe [21]. However, as this is not vital for the conceptual design of aircraft, it has not been looked into in detail. Future iterations should look into this as there is a need to ensure that the aircraft is manoeuvrable enough yet not overtly sensitive, guaranteeing the best possible handling qualities for the aircraft. This would not only allow the aircraft to be safe and easily certifiable, but also cut down the amount of time required to train pilots, thereby reducing the cost for airlines and making the aircraft more desirable.

5 Lateral Static Stability

The general yaw and roll stability criterion for any aircraft [9] is given by,

$$\frac{\partial C_n}{\partial \beta} \equiv C_{n,\beta} > 0 \quad (33)$$

$$\frac{\partial C_l}{\partial \beta} \equiv C_{l,\beta} < 0 \quad (34)$$

respectively. This must be ensured for the aircraft and it must be able to achieve trim in accordance

to the requirements from CS-25. Table 8 shows that the aircraft fulfils the general yaw and roll stability criterion.

Table 8: Values of lateral static stability derivatives

	$C_{n,\beta}$	$C_{l,\beta}$
Cruise	0.1726	-0.1302

5.1 Lateral Trim Requirements

Cruise conditions and symmetric thrust have been assumed for the following analysis. For a steady altitude, heading and airspeed, the generalised small-angle lateral trim requirement is given by [9],

$$\begin{bmatrix} C_{Y,\beta} & C_{Y,\delta_a} & C_{Y,\delta_r} \\ C_{l,\beta} & C_{l,\delta_a} & C_{l,\delta_r} \\ C_{n,\beta} & C_{n,\delta_a} & C_{n,\delta_r} \end{bmatrix} \begin{Bmatrix} \beta \\ \delta_a \\ \delta_r \end{Bmatrix} = - \begin{Bmatrix} (\Delta C_Y)_p + C_w \phi \\ (\Delta C_l)_p \\ (\Delta C_r)_p \end{Bmatrix} \quad (35)$$

where ΔY_p , Δl_p and Δn_p is the change in side force, roll moment and yaw moment respectively, due to the propulsion system at zero sideslip. The dimensionless forms are given by

$$(\Delta C_Y)_p \equiv \frac{\Delta Y_p}{qS_w}, (\Delta C_l)_p \equiv \frac{\Delta l_p}{qS_w b_w}, (\Delta C_n)_p \equiv \frac{\Delta n_p}{qS_w b_w}$$

Since symmetric thrust has been assumed, Δl_p is zero as propellers do not result in a rolling moment. $\Delta Y_p = 4.24$ kN and $\Delta n_p = 306$ Nm were found using the method outlined in Section 3. Equation 35 can be rearranged such that weight can be varied, allowing for the relationship between the total weight of the aircraft and the change in trim angles to be established as per Figure F.1. This shows that the most constraining case will be at the maximum take off weight. Hence, analysis was carried out for this case.

Figure 10 shows the angles required to achieve trim for a given bank angle. Since angles required are within those permissible by the aircraft, it has been concluded the aircraft can be trimmed in cruise.

Equation 35 can be recast to determine the required aileron and rudder deflection to achieve a constant sideslip and bank angle, in order to find the steady-heading sideslip (SHSS) as shown [9],

$$\begin{bmatrix} C_{Y,\delta_a} & C_{Y,\delta_r} & c_w \\ C_{l,\delta_a} & C_{l,\delta_r} & 0 \\ C_{n,\delta_a} & C_{n,\delta_r} & 0 \end{bmatrix} \begin{Bmatrix} \delta_a \\ \delta_r \\ \phi \end{Bmatrix} = - \begin{Bmatrix} C_{Y,\beta} \\ C_{l,\beta} \\ C_{n,\beta} \end{Bmatrix} \beta \quad (36)$$

The system of equations in Equation 36 can be solved by Gaussian elimination to obtain the following linear relations [9],

$$\begin{Bmatrix} \delta_a \\ \delta_r \\ \phi \end{Bmatrix} = \begin{Bmatrix} \left(\frac{\partial \delta_a}{\partial \beta} \right)_{SHSS} \\ \left(\frac{\partial \delta_r}{\partial \beta} \right)_{SHSS} \\ \left(\frac{\partial \phi}{\partial \beta} \right)_{SHSS} \end{Bmatrix} \beta \quad (37)$$

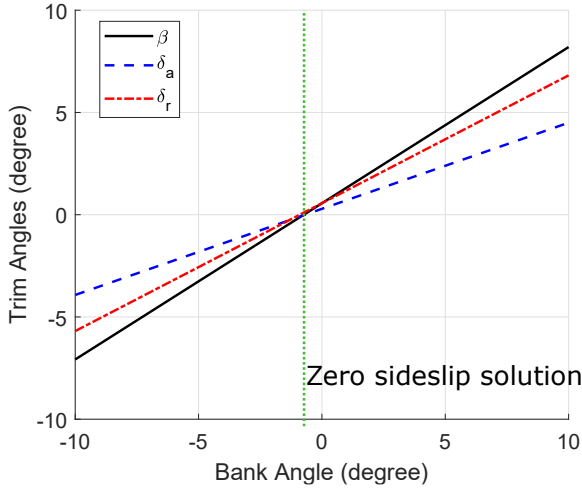


Figure 10: Relationship between sideslip angle, aileron deflection and rudder deflection required to achieve trim for a given bank angle.

where

$$\begin{aligned} \left(\frac{\partial \delta_a}{\partial \beta}\right)_{SHSS} &= \frac{C_{l,\delta_r} C_{n,\beta} - C_{l,\beta} C_{n,\delta_r}}{C_{l,\delta_a} C_{n,\delta_r} - C_{l,\delta_r} C_{n,\delta_a}} \\ \left(\frac{\partial \delta_r}{\partial \beta}\right)_{SHSS} &= \frac{C_{l,\beta} C_{n,\delta_a} - C_{l,\delta_a} C_{n,\beta}}{C_{l,\delta_a} C_{n,\delta_r} - C_{l,\delta_r} C_{n,\delta_a}} \\ \left(\frac{\partial \phi}{\partial \beta}\right)_{SHSS} &= -\frac{1}{c_w} \left[C_{Y,\delta_a} \left(\frac{\partial \delta_a}{\partial \beta}\right)_{SHSS} + C_{Y,\delta_r} \left(\frac{\partial \delta_r}{\partial \beta}\right)_{SHSS} + C_{Y,\beta} \right] \end{aligned}$$

Figure 11 shows the angles required to achieve trim for SHSS. Since angles required are within those permissible by the aircraft, it has been concluded that the aircraft is trimmable in this mode during cruise.

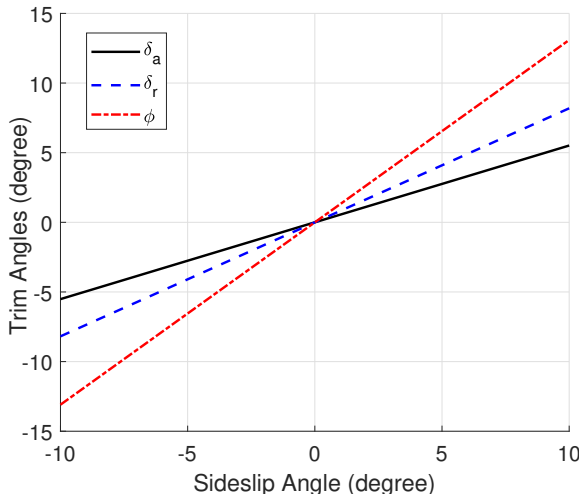


Figure 11: Relationship between trim angles required for a given sideslip angle.

5.2 Engine Failure and Minimum Control Airspeed

5.2.1 Methodology

Equation 35 can be rearranged such that aileron and rudder deflection per unit weight coefficient are subjects of the linear system [9],

$$\begin{Bmatrix} \beta/C_w \\ \delta_a/C_w \\ \delta_r/C_w \end{Bmatrix} = \begin{bmatrix} C_{Y,\beta} & C_{Y,\delta_a} & C_{Y,\delta_r} \\ C_{l,\beta} & C_{l,\delta_a} & C_{l,\delta_r} \\ C_{n,\beta} & C_{n,\delta_a} & C_{n,\delta_r} \end{bmatrix}^{-1} \begin{Bmatrix} -\frac{\Delta Y_p}{W} - \phi \\ -\frac{\Delta l_p}{b_w W} \\ -\frac{\Delta n_p}{b_w W} \end{Bmatrix} \quad (38)$$

After solving the system of equations in 38 for δ_a/C_w and δ_r/C_w and the required sideslip angle, values are used in equations 39 and 40 to find the aileron and rudder limited minimum control airspeed for a given configuration.

$$(V_{mc})_{aileron} = \sqrt{\frac{2W(\delta_a/C_w)}{\rho S_w \delta_{a,sat}}} \quad (39)$$

$$(V_{mc})_{rudder} = \sqrt{\frac{2W(\delta_r/C_w)}{\rho S_w \delta_{r,sat}}} \quad (40)$$

where $\delta_{a,sat} = \delta_{r,sat} = 25^\circ$ as they are the maximum deflections used in control surface sizing [27]. The CG position considered was the aftmost CG position as it would result in the largest yawing moment due to the side force generated by the propellers.

The main assumptions made to determine the minimum control airspeed are as follows:

1. No additional drag is produced when propellers are stationary or windmilling.
2. If a turboprop fails and it is not at full power, the remaining turboprop is throttled to full power and the additional power is split equally between the remaining electric motors.
3. If an electric motor fails, additional power is split between remaining electric motors. If this exceeds the maximum power rating on the electric motors, excess power goes to the turboprop.
4. At take off, turboprops are already at full power. The worst possible case is that power to the entire wing is lost. However, this is highly unlikely due to the presence of redundancies. The most realistic worst case scenario is that both turboprop and outboard electric motor fail while at maximum thrust.
5. Change in side force and yawing moment can be found using the method outlined in Section 3. Change in rolling moment is elaborated on below.

The change in rolling moment was approximated by assuming loss in lift, due to engine failure, is due to the loss in additional lift due to propeller effect. Since the total increase in lift coefficient due to the pro-

propeller effect was found to be $\Delta C_{L_{TO}} = 0.6719$ and $\Delta C_{L_{cruise}} = 0.0080$ [26][30], the contribution from each engine can be found by further assuming that the change in lift due to propeller effect is equal to the proportion of thrust produced. By considering momentum gain in the wake of the propeller, velocity of the flow aft of the propeller can be estimated. This value was found by Reference [31]. Differences between the free stream has been consolidated into Table 9. By considering the definition of the coefficient of lift, change in lift can be estimated by,

$$\Delta L_B = 0.25\rho\Delta V_B^2 S_w \Delta C_L \gamma \quad (41)$$

$$\Delta L_S = 0.125\rho\Delta V_S^2 S_w \Delta C_L (1 - \gamma) \quad (42)$$

The moment arm for the rolling moment has been assumed to the horizontal distance between the CG (along the span of the wing) and the centre of the propeller. The free body diagrams and equations used are found in Appendix G.

Table 9: The difference in velocity between the free stream flow and the flow aft of the propeller

	Take off	Cruise
Electric Motor	9.35 m/s	6.59 m/s
Turboprop	22.7 m/s	9.98 m/s

5.2.2 Results and Discussion

The absolute minimum control airspeed for a given failure case was found by solving Equation 38 and substituting results into Equations 39 and 40 for a range of bank angles to produce plots as seen in Figure 12. The absolute minimum control airspeed is obtained by finding the lowest minimum control airspeed in the feasible region.

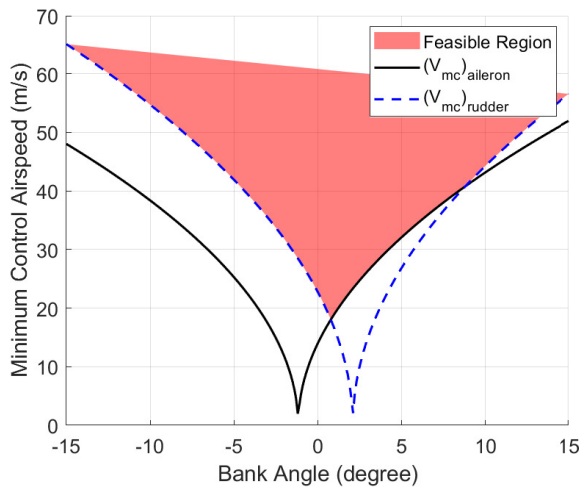


Figure 12: Minimum control airspeed against bank angle for take off with the outboard electric motor inoperative.

It was found that failure of an engine on the left wing is more critical. Considering that turboprops rotate

clockwise, the aircraft has a natural tendency to bank and head left. This was further verified as the minimum control airspeed graphs did not have a zero minimum control airspeed (with reference to Figures H.1 H.2) at a zero bank angle as expected [9].

Cruise, take off and landing condition were considered. However, it was found that engine failure during landing was relatively insignificant as engines are idling, thus not producing notable amounts of thrust. Take off was found to be the most constraining with absolute minimum control being higher than that at cruise conditions, as seen in Table 10. This has been attributed all engines producing maximum power at take off. Therefore, loss of an engine will result in the greatest loss of thrust and lift relative to conditions where all engines are operative.

Table 10: Minimum control airspeed for cruise and take off configurations.

	Cruise	Take off
Symmetric Thrust	2.33 m/s	3.21 m/s
Outboard Electric Motor Failure	17.7 m/s	17.8 m/s
Turboprop Failure	5.93 m/s	19.9 m/s
Two Critical Engines Failure	18.4 m/s	27.4 m/s

In accordance with CS-25.149 (b), the aircraft must be able to “maintain straight flight with an angle of bank of not more than 5°”. In the most constraining case shown in Figure 12, the maximum bank angle required at the absolute minimum control airspeed is less than 5°, illustrating the fulfilment of this criterion.

CS-25.149 (c) states all the conditions in which the minimum control airspeed needs to be considered to ensure the aircraft is certifiable. However, due to time constraint, all except CS-25.149 (c) (6) and (7) (requires the ground effect and drag of the propeller to be examined) have been considered. This is an area that should be looked into in future iterations.

The estimation of forces induced on the aircraft due to engine failure are simplistic. Firstly, it does not consider the coupling between roll and yaw. Secondly, the assumptions made to quantify the rolling moment generated by loss of an engine do not fully capture the physics and is merely a rough estimation of the moment generated. Lastly, the exact power split during these failure cases will provide the most precise estimate for the aircraft. However, these have not been determined by the propulsion team as aforementioned assumptions on the distribution of power were made.

Ergo, this is a major area to improve upon the fidelity of in the next iteration.

5.3 Rudder Sizing

Equation 35 can be rearranged to the form below to determine if rudder sizing is suitable.

$$\begin{Bmatrix} \beta \\ \delta_a \\ C_{n,\delta_r} \end{Bmatrix} = - \begin{bmatrix} C_{Y,\beta} & C_{Y,\delta_a} & C_{Y,\delta_r} \\ C_{l,\beta} & C_{l,\delta_a} & C_{l,\delta_r} \\ C_{n,\beta} & C_{n,\delta_a} & C_{n,\delta_r} \end{bmatrix}^{-1} \begin{Bmatrix} (\Delta C_Y)_p + C_w \phi \\ (\Delta C_l)_p \\ (\Delta C_r)_p \end{Bmatrix} \quad (43)$$

It was found that rudder sizing by Reference [27] was suitable. Using an analogous method to verify the aileron sizing should be looked into in future iterations.

6 Aileron Propeller Interaction

A preliminary study showed no evidence in literature to support the use of control surface that does not experience free stream conditions (i.e. an aileron behind a propeller). Thus, a design decision was made to constrain propulsion and aerodynamics teams to only use 20 – 40% of the wing for their purposes. Remaining space must be left for ailerons. Upon further reading of literature regarding conceptual aircraft with distributed propulsion, three main schools of thought were found.

Firstly, keeping ailerons in clean flow. This was explored by References [32] and [33]. The rationale behind this was not elaborated upon. In addition, there is little information within literature to suggest methodology to quantify the impacts of this compromise. The main disadvantage is that benefits of distributed propulsion cannot be fully exploited. However, conventional methods can be used to size control surfaces.

Secondly, complete replacement of control surfaces by the distributed propulsion system where yaw control is attained by asymmetric thrust and pitch and roll control through a combination of thrust vectoring and asymmetric thrust. This eliminates the need for empennage, resulting in weight saving [34]. However, this is not possible due to safety regulations which would dictate the need for primary control surfaces as a form of redundancies, resulting in unnecessary weight and cost penalty.

If the architecture of the propulsion system remains the same, electric motors draw power directly from the turboprops during cruise. Therefore, the total amount of thrust needs to be increased in order to achieve a given change in the roll, pitch or yaw. The time constant of the whole system is dependent on the turboprop and not on electrical motors which have significantly shorter time constants [35]. This limits how fast a change can be achieved, thus making the aircraft potentially unsafe given flight critical situations: the ability to throttle and change power output is still limited by the turboprop.

In addition, if the architecture remains the same and electric motors are used as control surfaces, it would

not be possible to certify the aircraft based on CS-25.143 (B)(2) which requires that the aircraft is still controllable given “the sudden failure of the second critical engine” [29]. Since a critical engine is defined as one “whose failure would most adversely affect the performance or handling qualities of an aircraft” [29], the second turboprop is the second critical engine. Failure of this would result in the aircraft experiencing an almost complete power loss and hence, loss of controls. Thus, the aircraft will not be certifiable. If thrust vectoring is to be adopted, one of the following would potentially be required: a change in propulsion architecture, a secondary power generator which could take the form of an auxiliary power unit (APU) to generate power or carry sufficient excess batteries for electric motors in the event of failure of the second critical engine. Therefore, the use of thrust vectoring to replace control surfaces will not be a viable option.

Lastly, placement of ailerons behind the propeller. This is currently being tested by NASA’s Hybrid Electric Integrated Systems Testbed (HEIST) research into distributed propulsion whereby the X-57 concept aircraft has ailerons behind the propellers. However, there is limited information on the impact on the consequences of this design choice. Since it is not possible to estimate at what aileron deflection angle flow will separate (due to the lack of data) and thus lose control of the aircraft, it is not possible to size ailerons, guarantee the safety of the passengers and certify the aircraft. However, this will be a good area to look into in the next iteration with unsteady computational fluid dynamics (CFD) simulations and physical tests to determine the feasibility and sizing required. In conclusion, given the timescale of the project, the first school of thought was adopted.

7 Conclusion

Since the aircraft is stable and both the static analysis and dynamic analysis carried out by [1] are in agreement, the aircraft is indeed stable. There is also agreement between the initial tail and control sizing. It is likely that the reason behind accurate initial estimates is due to the presence of historical data and a conventional aircraft configuration. In terms of static stability, the impact of hybridisation is negligible except in terms of the ailerons.

Although the configuration was found to be statically stable both laterally and longitudinally, it is far from being optimal and there are multiple areas that could be improved and looked into. Given more time and a chance to rectify the issues, there are multiple changes that would be made. For example, additional failure modes of the empennage such as T-tail flutter should be looked into and developing a method to quantify design decisions in terms of cost to ensure the aircraft is desirable to buyers.

References

- [1] I. Karatsivoulis. Hybrid electric transport fatt 05 flight dynamics and control individual report. Completed as part of AE3-306 Group Design Project, June 2019.
- [2] W.B. Kemp. Definition and application of longitudinal stability derivatives for elastic airplanes. Technical report, NASA Langley Research Center, 1972. Available online at: <https://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/19720012356.pdf> (last accessed 15/06/2019).
- [3] A.D. Ronch and et al. Benchmarking ceasim software to predict flight controls and flying qualities of the b-747. *27th International Congress of the Aeronautical Sciences*, 2010.
- [4] M. Kuitche and R. Botez. Methodology of estimation of aerodynamic coefficients of the uas4 ethical datcom and vlm procedure. *AIAA Modeling and Simulation Technologies Conference*, 2017. Available online at: <https://arc.aiaa.org/doi/abs/10.2514/6.2017-3152> (last accessed 15/06/2019).
- [5] T. Melin. A vortex lattice matlab implementation for linear aerodynamic wing applications. Master thesis, Royal Institute of Technology (KTH), December 2000. Available online at: https://www.researchgate.net/publication/320924804_A_Vortex_Lattice_MATLAB_Implementation_for_Linear_Aerodynamic_Wing_Applications (last accessed 15/06/2019).
- [6] T. Melin. *User's Guide and Reference Manual for Tornado*. Royal Institute of Technology (KTH).
- [7] J. Roskam. *Airplane Design*. Roskam Aviation and Engineering Corporation, 1985.
- [8] McDonnell Douglas Astronautics Company. *The USAF Stability and Control DATCOM*, 1979.
- [9] W.F. Phillips. *Mechanics of Flight*. John Wiley & Sons, second edition, 2010.
- [10] R.R. Tanner and T.D. Montgomery. Stability and control derivative estimates nasa-tm-72863, January 1979. Available online at: https://www.esdu.com/cgi-bin/ps.pl?sess=unlicensed_1190615164833pcb&t=doc&p=nasa_tm72863 (last accessed 15/06/2019).
- [11] R.C. Nelson. *Flight Stability and Automatic Control*. McGraw-Hill, second edition, 1998.
- [12] R.F. Stengel. *Flight Dynamics*. Princeton University Press, first edition, 2004.
- [13] B.W. McCormick. *Aerodynamics, Aeronautics and Flight Mechanics*. John Wiley & Sons, second edition, 1995.
- [14] R. Garcia-arduengo. Hybrid electric transport fatt 02 propulsion systems. Completed as part of AE3-306 Group Design Project, June 2019.
- [15] H. Ribner. Notes on the propeller and slipstream in relation to stability. National Advisory Committee for Aeronautics Wartime Report, October 1944. Available online at: <https://ntrs.nasa.gov/search.jsp?R=19930092551> (last accessed 15/06/2019).
- [16] W. Phillips, E. Anderson, and Q. Kelly. Predicting the contribution of running propellers to aircraft stability derivatives. *Journal of Aircraft*, 40(6):1107–1114, December 2003. Available online at: <https://arc.aiaa.org/doi/pdf/10.2514/2.7221> (last accessed 15/06/2019).
- [17] R. Rooyen and M. Eshelby. Assesment of propeller influence on lateral-directional stability of multi-engine aircraft. *Journal of Aircraft*, 18(5):364–371, May 1981. Available online at: <https://arc.aiaa.org/doi/10.2514/3.44710> (last accessed 15/06/2019).
- [18] H. Ribner. Propellers in yaw. National Advisory Committee for Aeronautics Wartime Report, December 1943. Available online at: <https://apps.dtic.mil/dtic/tr/fulltext/u2/a801088.pdf> (last accessed 15/06/2019).
- [19] H. Ribner. Formulas for propellers in yaw and charts of the side-force derivatives. National Advisory Committee for Aeronautics Wartime Report, May 1943. Available online at: <https://apps.dtic.mil/dtic/tr/fulltext/u2/a801215.pdf> (last accessed 15/06/2019).
- [20] H. Ribner. Proposal for a propeller side-force factor. National Advisory Committee for Aeronautics Wartime Report, December 1943. Available online at: <http://naca.central.cranfield.ac.uk/reports/1943/naca-wr-1-336.pdf> (last accessed 15/06/2019).
- [21] B. Etkin & L.D. Reid. *Dynamics of Flight - Stability and Control*. John Wiley & Sons, third edition, 1996.
- [22] A. Lau. Hybrid electric transport fatt 01 configuration defintion. Completed as part of AE3-306 Group Design Project, June 2019.
- [23] B. Hoon. Hybrid electric transport fatt 01 configuration defintion. Completed as part of AE3-306 Group Design Project, June 2019.

- [24] D P Raymer. *Aircraft Design: A Conceptual Approach*. American Institute of Aeronautics and Astronautics, fifth edition, 2012.
- [25] D.K. Schmidt. *Modern Flight Dynamics*. McGraw-Hill, international edition, 2012.
- [26] Z. Zareer. Hybrid electric transport fatt 03 aerodynamics individual report. Completed as part of AE3-306 Group Design Project, June 2019.
- [27] Y.J. Ho. Hybrid electric transport fatt 05 flight dynamics and control individual report. Completed as part of AE3-306 Group Design Project, June 2019.
- [28] H. Ashley. *Engineering Analysis of Flight Vehicles*. Dover Publications Inc, new edition, May 2012.
- [29] European Aviation Safety Agency. Certification specifications for large aeroplanes cs-25, 2016. Available online at: https://www.easa.europa.eu/sites/default/files/dfu/CS-25%20Amendment%2016%20-%20consolidated%20version%20for%20information%20only_0.pdf (last accessed 04/12/2018).
- [30] M. Hadjiandreou. Hybrid electric transport fatt 03 aerodynamics. Completed as part of AE3-306 Group Design Project, June 2019.
- [31] B. Mao. Hybrid electric transport fatt 02 propulsion systems. Completed as part of AE3-306 Group Design Project, June 2019.
- [32] J. Hermetz, M. Ridel, and C. Doll. Distributed electric propulsion for small business aircraft a concept-plane for key-technologies investigations. *30th Congress of the International Council of the Aeronautical Sciences*, 2016. Available online at: <https://hal.archives-ouvertes.fr/hal-01408988/document> (last accessed 16/06/2019).
- [33] Onera. Pioneering concepts for personal air transport systems. Available online at: <http://www.nianet.org/ODM/ODM%20Wednesday%20presentations%20Final/13%20LeTallec%20ONERA.pdf> (last accessed 16/06/2019).
- [34] G. Ameyugo, M. Taylor, and R. Singh. Distributed propulsion feasibility studies. Technical report, Cranfield University, 2006. Available online at: http://icas.org/ICAS_ARCHIVE/ICAS2006/PAPERS/250.PDF (last accessed 08/06/2019).
- [35] H.D. Kim, A.T. Perry, and P.J. Ansell. A review of distributed electric propulsion concepts for air vehicle technology. Technical report, NASA Armstrong Flight Research Center, 2018. Available online at: <https://www1.grc.nasa.gov/wp-content/uploads/2018EATS-Review-of-DEP-Hyun-D.-6.2018-4998.pdf> (last accessed 14/06/2019).

A Tornado Model

Figures A.1, A.2, A.3 and A.4 show the model (made by Reference [1] and the aerodynamics team) used in Tornado to obtain the stability derivatives.

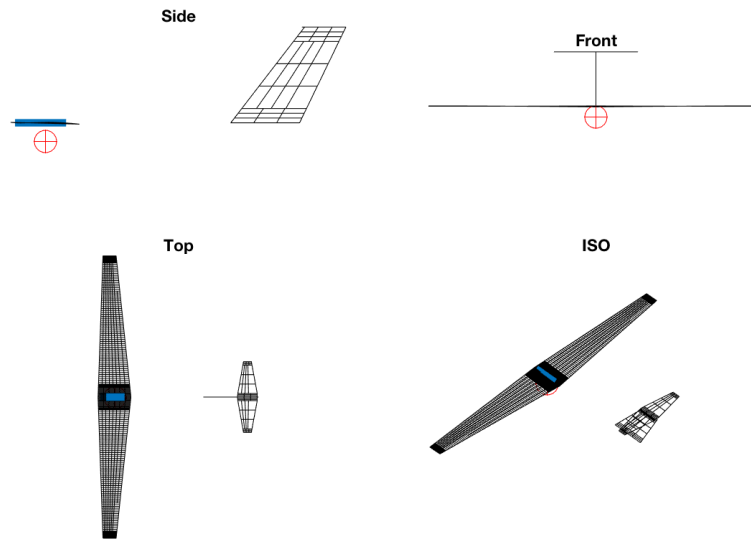


Figure A.1: 3 view of Tornado model [1]



Figure A.2: Wing and empennage partition layout [1]

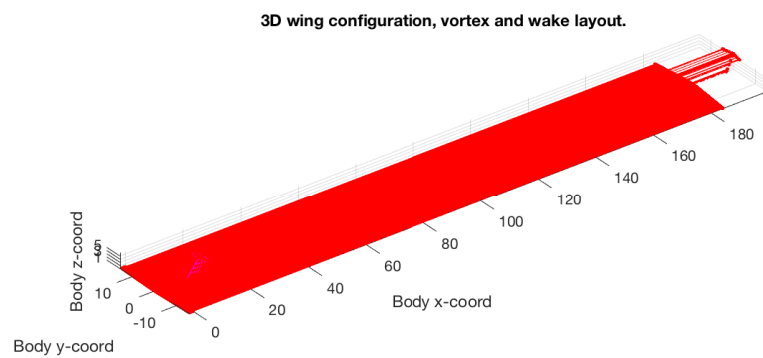


Figure A.3: Wing configuration of Tornado model [1]

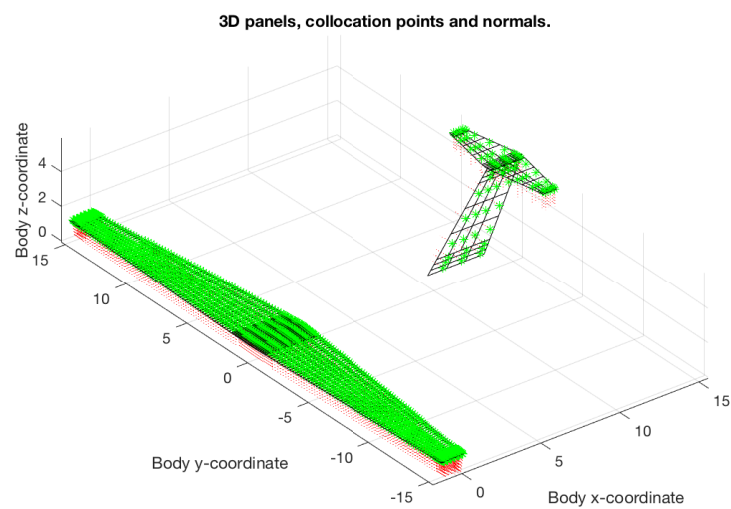


Figure A.4: 3D panels, collocation points and normals of Tornado model [1]

B Isolated Control Surfaces Plot

The following plots were used as a sanity check to ensure that the stability derivatives obtained through Tornado and empirical estimates were within the expected order of magnitude as mentioned in Subsection 2.2.

Figures B.1 and B.2 are plots of Equation 2 for a variation in angle of attack and elevator deflection.

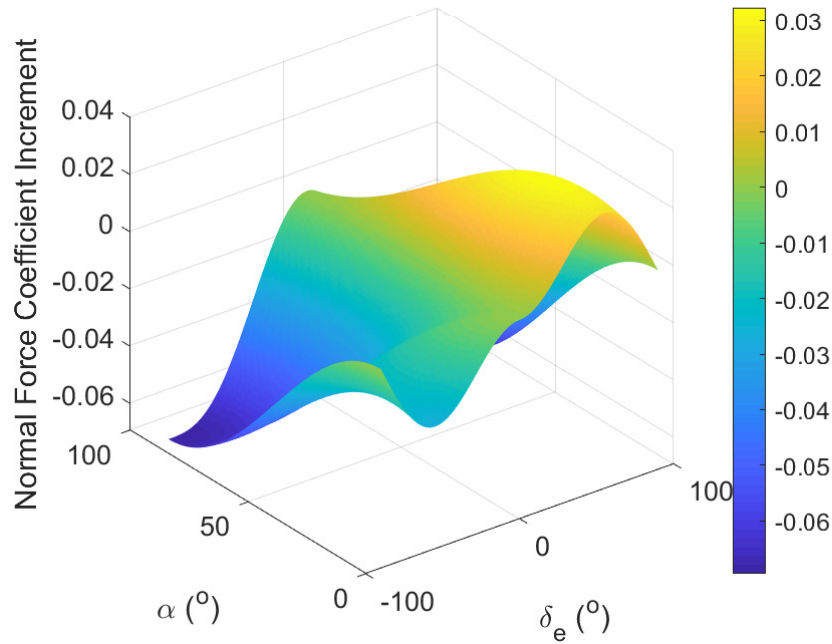


Figure B.1: 3D plot of normal force increment due to an elevator deflection and change in angle of attack

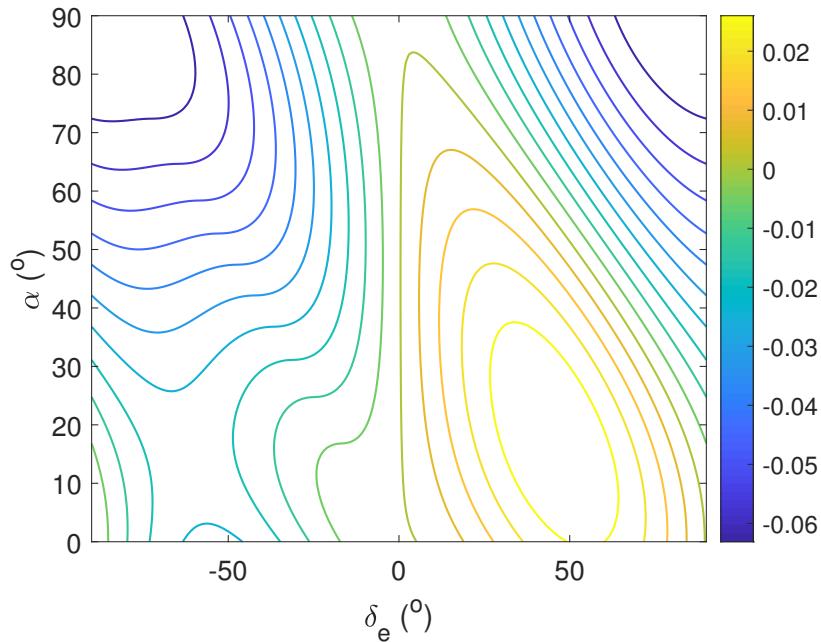


Figure B.2: Contour plot of normal force increment due to an elevator deflection and change in angle of attack

Figures B.3 and B.4 are plots of Equation 3 for a variation in angle of attack and elevator deflection.

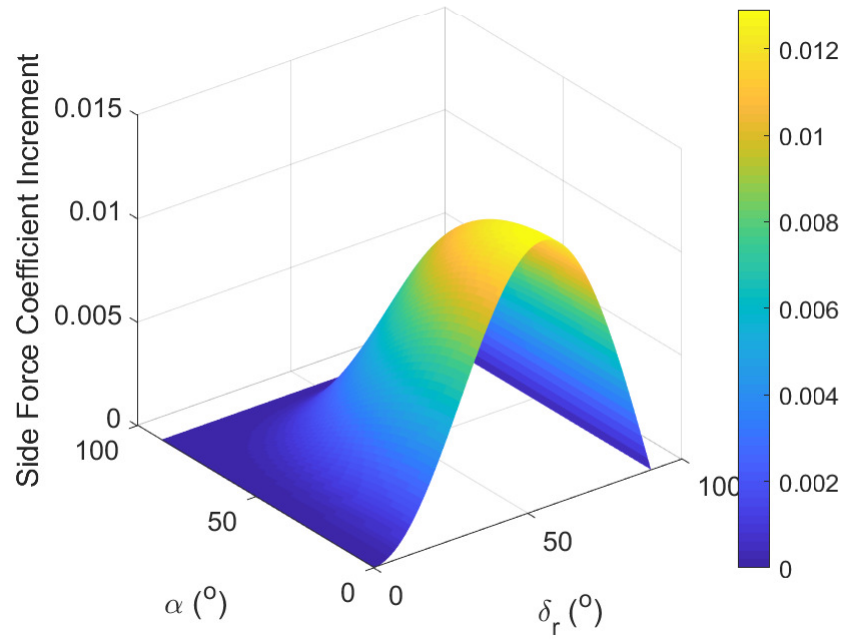


Figure B.3: 3D plot of side force increment due to a rudder deflection and change in angle of attack

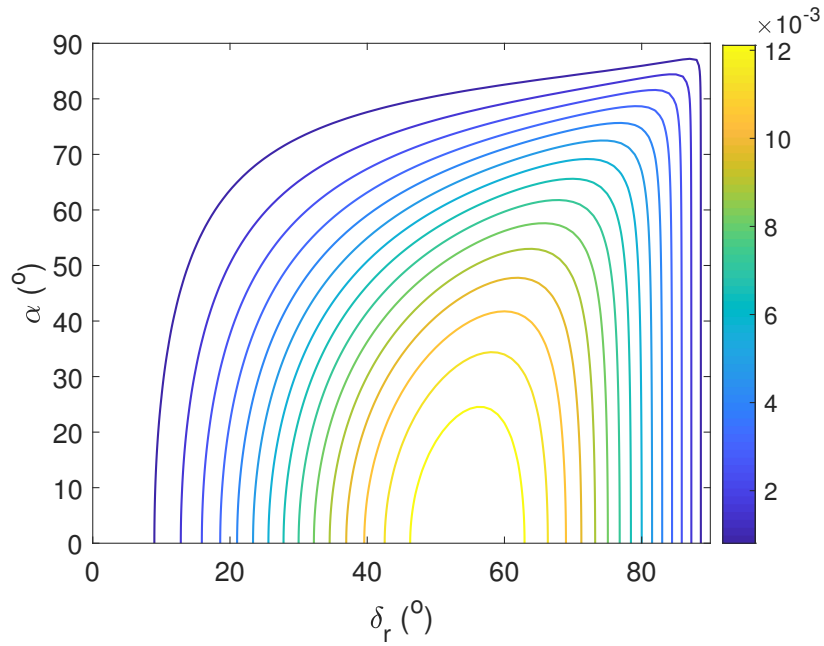


Figure B.4: Contour plot of side force increment due to a rudder deflection and change in angle of attack

Figures B.5 and B.6 are plots of Equation 5 for a variation in angle of attack and elevator deflection.

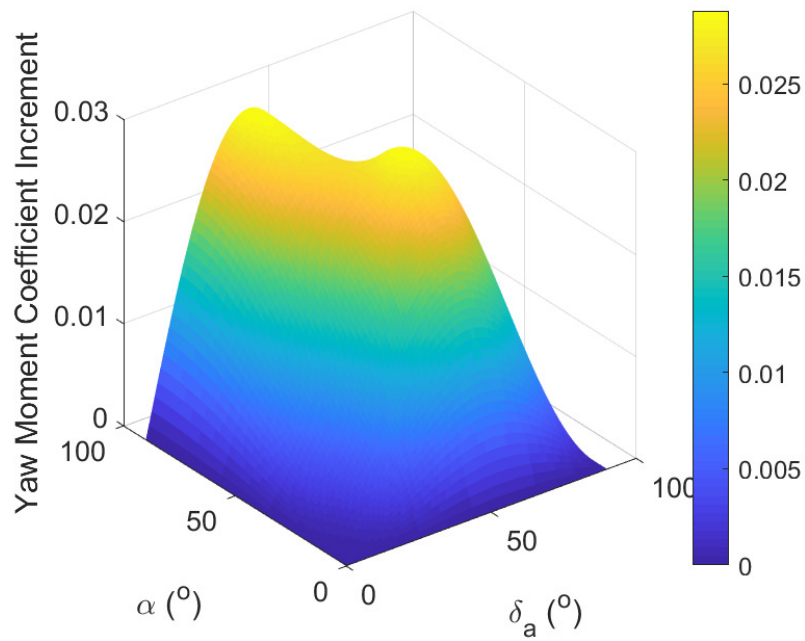


Figure B.5: 3D plot of yaw moment increment due to an aileron deflection and change in angle of attack

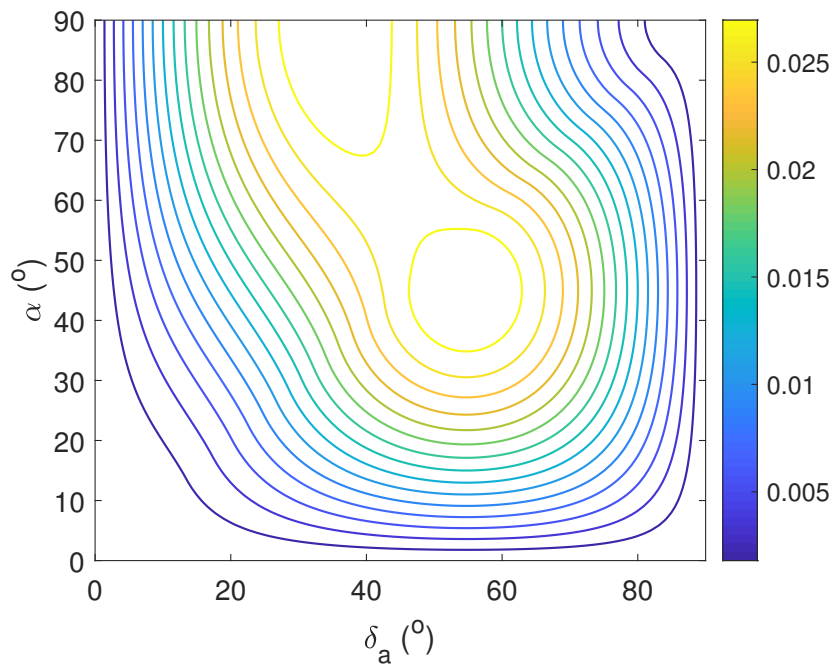


Figure B.6: Contour plot of yaw moment increment due to an aileron deflection and change in angle of attack

C Derivation for Propellers Correction

C.1 Pitching Moment

Assumptions:

1. All turboprops and electric motors produce thrust and normal force in the same direction
2. The aircraft is symmetric about the xz plane
3. Small thrust angles such that it can be assumed that thrust is equal to the drag

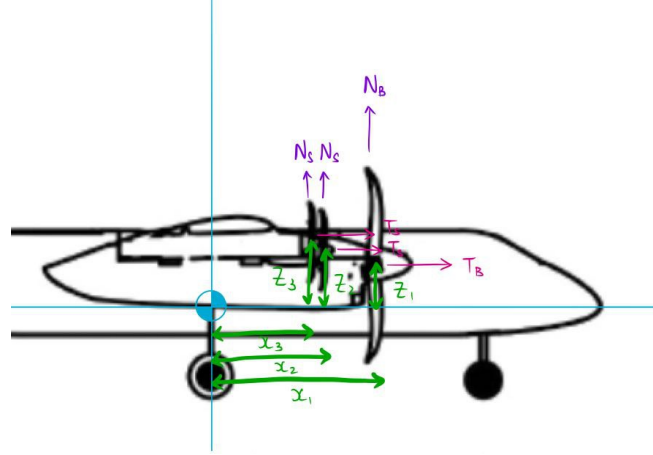


Figure C.1: Free body diagram for effect of propellers on pitching moment

Taking moments about the CG where a positive moment results in a pitching up moment,

$$(m_{cg})_p = 2N_S x_3 + 2N_S x_2 + 2N_B x_1 - 2T_S z_3 - 2T_S z_2 - 2T_B z_1$$

The dimensionless change in pitching moment due to the propellers are given by

$$\begin{aligned} (\Delta C_m)_p &= \frac{(m_{cg})_p}{0.5\rho V_\infty^2 S_w \bar{c}_w} \\ &= 2 \left\{ \frac{x_3}{\bar{c}_w} \frac{N_S}{0.5\rho V_\infty^2 S_w} + \frac{x_2}{\bar{c}_w} \frac{N_S}{0.5\rho V_\infty^2 S_w} + \frac{x_1}{\bar{c}_w} \frac{N_B}{0.5\rho V_\infty^2 S_w} - \frac{z_3}{\bar{c}_w} \frac{T_S}{0.5\rho V_\infty^2 S_w} - \frac{z_2}{\bar{c}_w} \frac{T_S}{0.5\rho V_\infty^2 S_w} - \frac{z_1}{\bar{c}_w} \frac{T_B}{0.5\rho V_\infty^2 S_w} \right\} \\ &= 2 \left\{ \frac{2d_{ps}^2}{S_w \bar{c}_w J_S^2} C_{N_{S,\alpha}} (x_2 + x_3) \alpha_p + \frac{2d_{ps}^2}{S_w \bar{c}_w J_S^2} C_{N_{S,\alpha}} x_1 \alpha_p - \frac{(1-\gamma)(z_2 + z_3)}{4\bar{c}_w} C_d - \frac{z_1 \gamma}{2\bar{c}_w} C_d \right\} \end{aligned}$$

where ε_{dp} is the local downwash angle at the propeller, α_{0p} is the thrust setting angle of the aircraft and since $\alpha_p = \alpha - \varepsilon_{dp} + \alpha_{0p}$ the above can be rearranged into the form seen in Equation 16.

C.2 Yawing Moment

Assumptions:

1. Aircraft is symmetric about the xz plane
2. Both turboprops spin in the clockwise direction when viewed from behind the aircraft
3. The electric motors on the right wing spin clockwise when viewed from behind the aircraft
4. The electric motors on the left wing spin anti-clockwise when viewed from behind the aircraft
5. The axisymmetry of the propellers allows for the side force due to a given sideslip angle to be related in the same way as the normal force due to a given angle of attack [9]

$$C_{Y_p, \beta} = -C_{N_p, \alpha}$$

Taking moments about the centre of gravity assuming that a clockwise rotation about the z-axis is positive and that there is symmetric thrust,

$$\begin{aligned} (n_{cg})_p &= 2n_B + 2n_S - 2n_S + 2Y_B x_1 + 2Y_S x_2 + 2Y_S x_3 \\ &= 2n_B + 2Y_B x_1 + 2Y_S x_2 + 2Y_S x_3 \end{aligned}$$

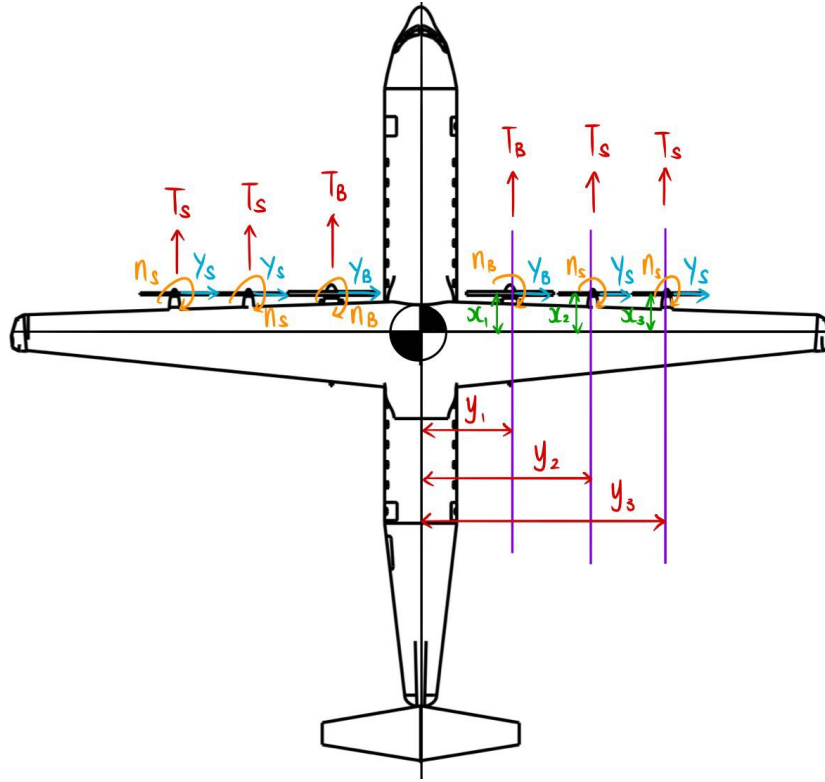


Figure C.2: Free body diagram for the effect of propellers on yawing moment.

The dimensionless change in yawing moment due to the propellers are given by

$$\begin{aligned}
 (\Delta C_n)_p &= \frac{(n_{cg})_p}{qS_w b_w} = 2 \left\{ \frac{n_B}{qS_w b_w} + \frac{Y_B}{qS_w b_w} x_1 + \frac{Y_S}{qS_w b_w} x_2 + \frac{Y_S}{qS_w b_w} x_3 \right\} \\
 &= 2 \left\{ \frac{\rho(\omega/2\pi)^2 d_{pB}^5}{qS_w b_w} C_{n_B} + \frac{\rho(\omega/2\pi)^2 d_{pB}^4}{qS_w b_w} C_{Y_B} x_1 + \frac{\rho(\omega/2\pi)^2 d_{pB}^4}{qS_w b_w} C_{Y_S} x_2 + \frac{\rho(\omega/2\pi)^2 d_{pB}^4}{qS_w b_w} C_{Y_S} x_3 \right\} \\
 &= 2 \left\{ \frac{2d_{pB}^3}{S_w b_w J_B^2} C_{n_{B,\alpha}} \alpha_p + \frac{2d_{pB}^2 x_1}{S_w b_w J_B^2} C_{N_{B,\alpha}} \beta_p + \frac{2d_{pS}^2}{S_w b_w J_S^2} (x_2 + x_3) C_{N_{S,\alpha}} \beta_p \right\}
 \end{aligned}$$

where $\beta_p = \beta - \varepsilon_{sp}$ and ε_{sp} is the local side wash angle at the location of the propeller. This allows for the above equation to be rearranged into the form seen in 17.

D Derivation for Stick Fixed Neutral Point

At the stick fixed neutral point, $\frac{\partial C_m}{\partial \alpha} = 0$, therefore,

$$-C_{L_{\alpha w}} \frac{x_{acw} - x_{cg}}{\bar{c}_w} + C_{M_{\alpha f}} - \eta_h C_{L_{\alpha h}} \left(1 - \frac{\partial \varepsilon}{\partial \alpha}\right) \frac{S_h}{S_w} \frac{x_{ach} - x_{cg}}{\bar{c}_w} + \frac{2d_{pS}^2 C_{N_{S,\alpha}}}{S_w \bar{c}_w J_S^2} (2x - x_{om} - x_{im}) \left(1 - \left(\frac{\partial \varepsilon}{\partial \alpha}\right)_p\right) + \frac{2d_{pB} C_{N_{B,\alpha}}}{S_w \bar{c}_w J_B^2} \left(\frac{\partial \varepsilon}{\partial \alpha}\right)_p (x - x_p) = 0 \quad (44)$$

where x_{om} is the distance along the x-axis (nose as reference point) to the outermost electric motor, x_{im} is the distance along the x-axis (nose as reference point) to the innermost electric motor and x_p is the distance along the x-axis (nose as reference point) to the turboprop.

Rearranging Equation 44 to make $\frac{x}{\bar{c}_w}$ to be the subject of the equation, results in the equation for the stick fixed neutral point and is given by,

$$\frac{x_{np}}{\bar{c}_w} = \frac{\frac{x_{acw}}{\bar{c}_w} C_{L_{\alpha w}} + C_{M_{\alpha f}} + C_{L_{\alpha w}} \eta_h \left(1 - \frac{\partial \varepsilon}{\partial \alpha}\right) \frac{S_h}{S_w} \frac{x_{ach}}{\bar{c}} + \frac{2d_{pS}^2 C_{N_{S,\alpha}}}{S_w \bar{c}_w J_S^2} (x_{om} + x_{im}) \left(1 - \frac{\partial \varepsilon}{\partial \alpha}\right)}{\xi} + \frac{\frac{2d_{pB} C_{N_{B,\alpha}}}{S_w \bar{c}_w J_B^2} \frac{\partial \varepsilon}{\partial \alpha} x_p}{\xi} \quad (45)$$

where

$$\xi = C_{L_{\alpha w}} + \eta_h \left(1 - \frac{\partial \varepsilon}{\partial \alpha}\right) \frac{S_h}{S_w} C_{L_{\alpha w}} + \frac{4d_{pS}^2 C_{N_{S,\alpha}}}{S_w J_s^2} \left(1 - \frac{\partial \varepsilon}{\partial \alpha}\right) + \frac{2d_{pB}^2 C_{N_{B,\alpha}}}{S_w J_s^2} \left(1 - \frac{\partial \varepsilon}{\partial \alpha}\right)$$

E Landing Trim Plot

Figure E.1 is the trim plot at landing and shows that the aircraft is able to achieve trim given a range of tail setting angle between -3° and 4°

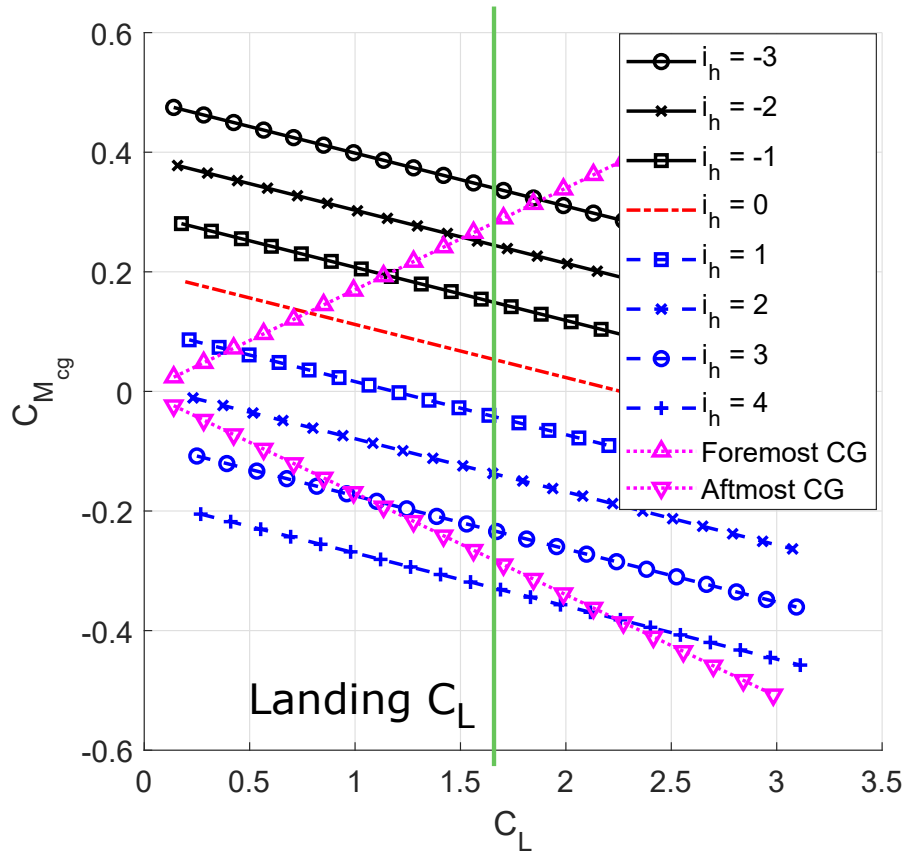


Figure E.1: Trim plot for landing at average CG location

F Variation of Lateral Trim Angles with Weight

Figure F.1 shows that the maximum take off weight will be the most constraining case for lateral trim.

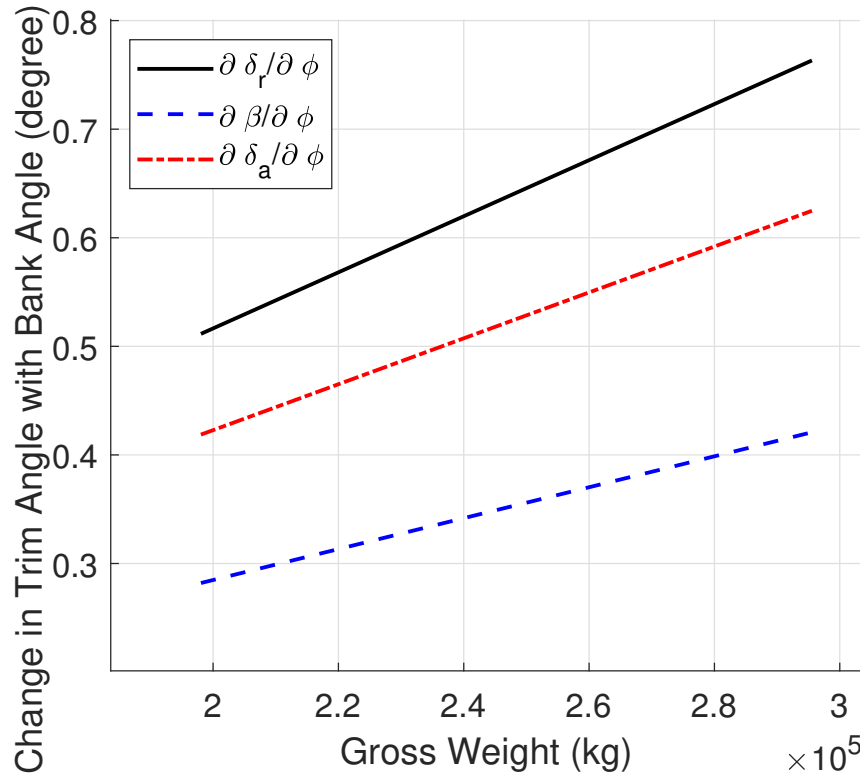


Figure F.1: Relationship between a change in gross weight and the change in trim angles with bank angle.

G Engine Failure Cases

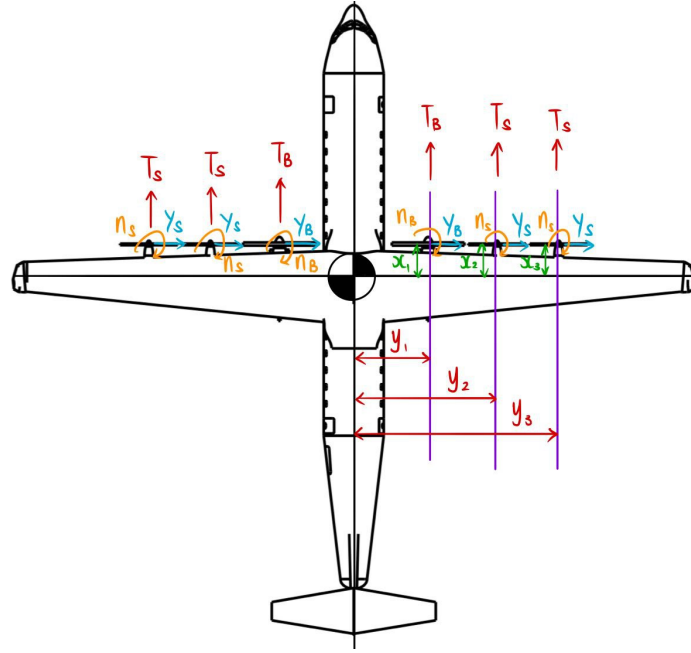


Figure G.1: Symmetric thrust - case 1

Equations obtained through force and moment equilibrium that correspond to case 1,

$$\Delta Y_p = 2Y_B + 4Y_S$$

$$\Delta n_p = 2n_B + 2Y_B x_1 + 2Y_S x_2 + 2Y_S x_3$$

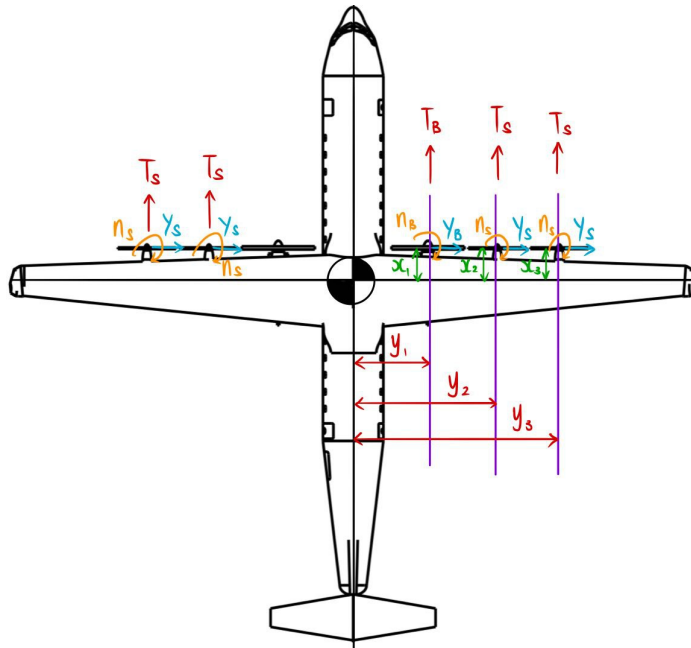


Figure G.2: Failure of turboprop on the left wing - case 2

Equations obtained through force and moment equilibrium that correspond to case 2,

$$\Delta Y_p = Y_B + 4Y_S$$

$$\Delta n_p = n_B + Y_B x_1 + 2Y_S x_2 + 2Y_S x_3 - T_B y_1$$

$$\Delta l_p = -\Delta L_B y_1$$

where y_{ac_w} is the aerodynamic centre of the wing along the y-axis. Assume that the loss of lift takes place at the centre of the engine.

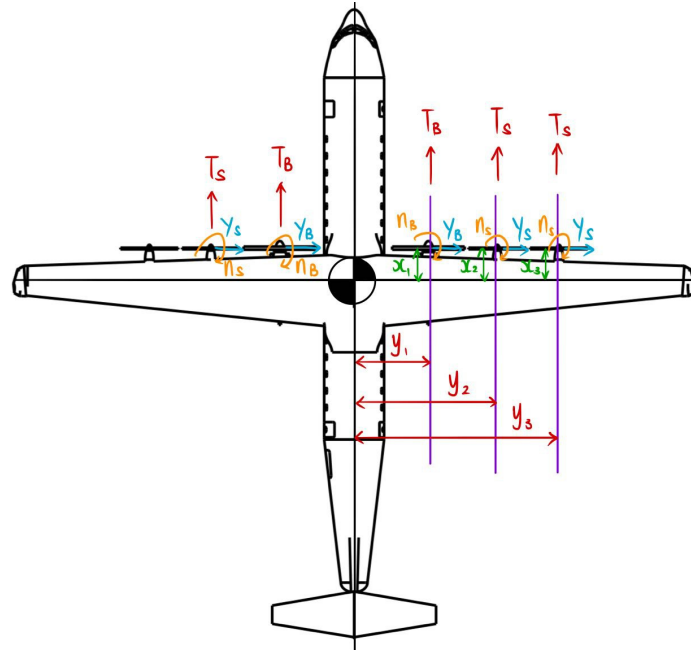


Figure G.3: Failure of the outermost electric motor - case 3

Equations obtained through force and moment equilibrium that correspond to case 3,

$$\begin{aligned}\Delta Y_p &= 2Y_B + 3Y_S \\ \Delta n_p &= 2n_B + n_s + 2Y_B x_1 + 2Y_S x_2 + Y_S x_3 - T_S y_3 \\ \Delta l_p &= -\Delta L_S y_3\end{aligned}$$

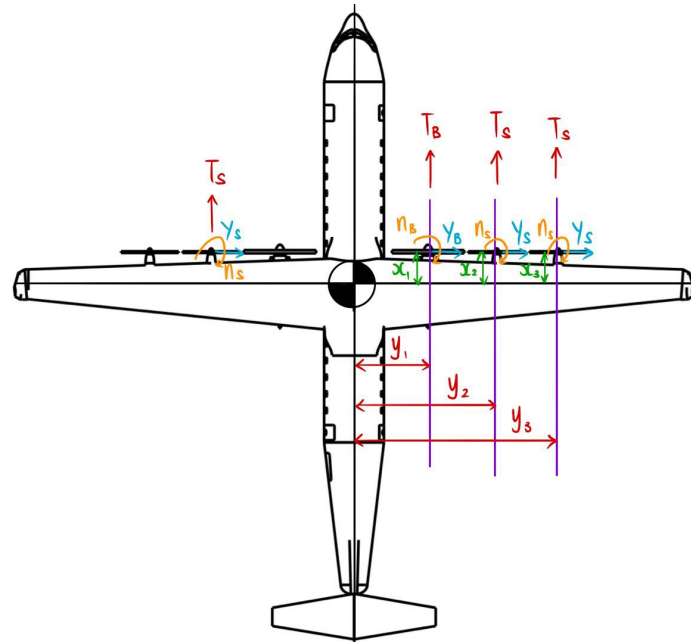


Figure G.4: Two critical engines inoperative - case 4

Equations obtained through force and moment equilibrium that correspond to case 4,

$$\begin{aligned}\Delta Y_p &= Y_B + 3Y_S \\ \Delta n_p &= n_B + n_s + Y_B x_1 + 2Y_S x_2 + Y_S x_3 - T_S y_3 - T_B y_1 \\ \Delta l_p &= -\Delta L_B y_1 - \Delta L_S y_3\end{aligned}$$

Failure of the left wing (when viewed from above) is more critical because the aircraft already has a tendency to roll and yaw to the left wing.

H Minimum Control Airspeed Graphs

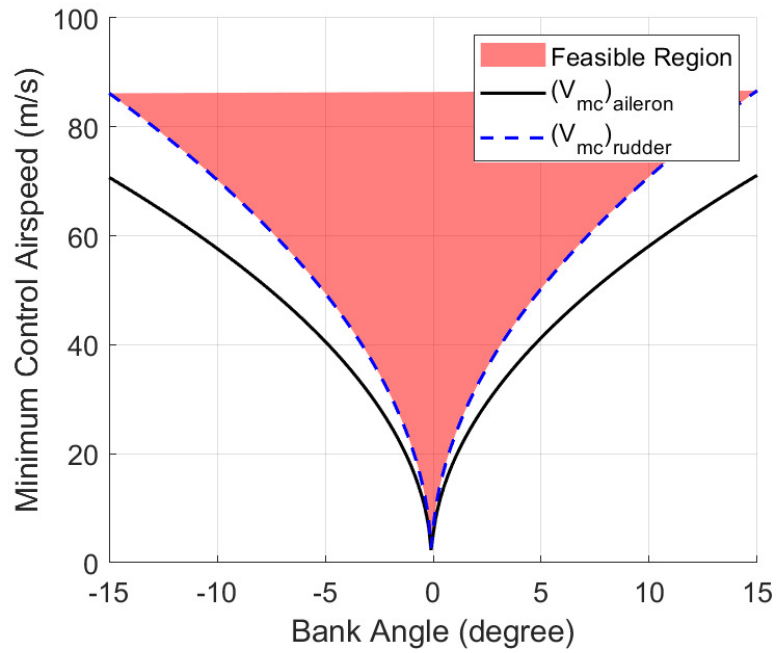


Figure H.1: Minimum control airspeed against bank angle in cruise conditions with symmetric thrust

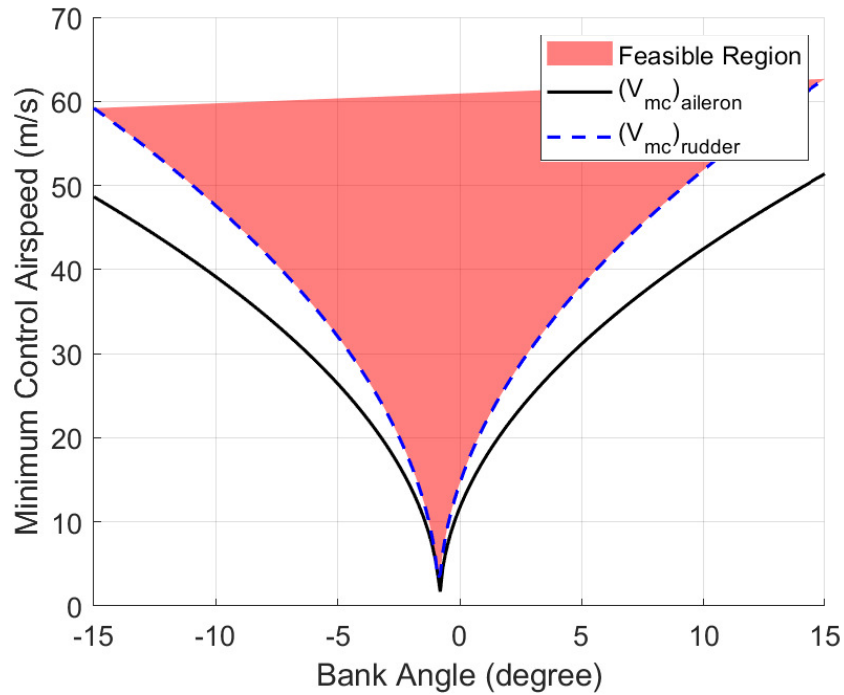


Figure H.2: Minimum control airspeed against bank angle in take off conditions with symmetric thrust

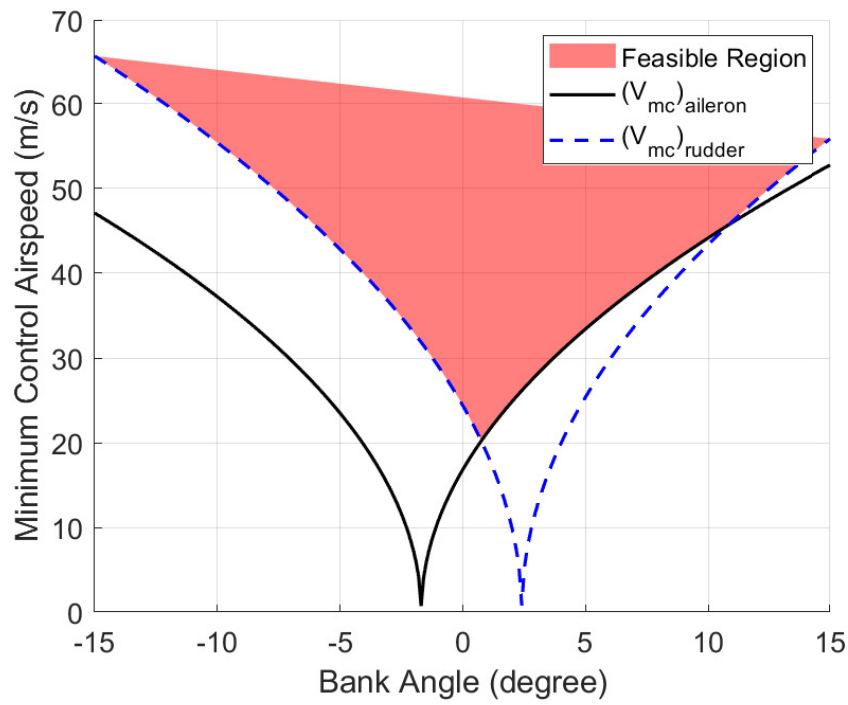


Figure H.3: Minimum control airspeed against bank angle in take off conditions with turboprop inoperative

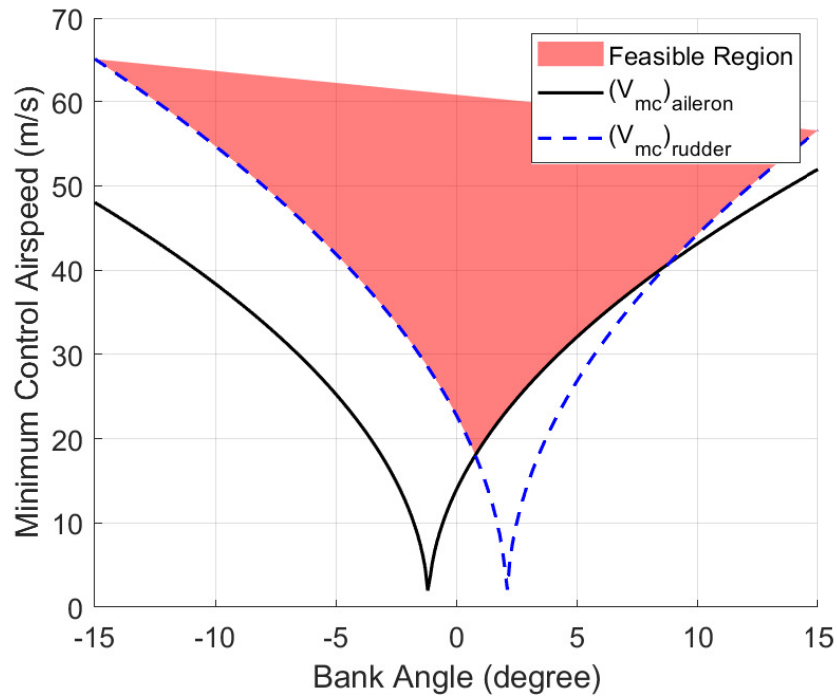


Figure H.4: Minimum control airspeed against bank angle in take off conditions with outboard electric motor inoperative